Instructor's Solutions Manual

Fundamental Principles of Optical Lithography
The Science of Microfabrication

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This manual contains solutions to all of the problems found in *Fundamental Principles of Optical Lithography: The Science of Microfabrication* by Chris Mack. The solutions were also worked out and provided in this format by Chris Mack. Any comments about these solutions, or about the problems in general, are welcome and should be sent to chris@lithoguru.com. Also, please check the errata page at www.lithoguru.com/textbook for any errata related to homework problems.

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Chapter 1

1.1. When etching an oxide contact hole with a given process, the etch selectivity compared to photoresist is found to be 2.5. If the oxide thickness to be etched is 140 nm and a 50% overetch is used (that is, the etch time is set to be 50% longer than that required to just etch through the nominal oxide thickness), what is the minimum possible photoresist thickness (that is, how much resist will be etched away)? For what reasons would you want the resist to be thicker than this minimum?

Let: R_{oxide} = oxide etch rate
R_{resist} = resist etch rate
t = time required to just etch the oxide, without overetch

Thus,

$$R_{oxide}\, t = 140\ nm \quad \text{and} \quad R_{resist} = \frac{R_{oxide}}{2.5}$$

For the 50% overetch, the etch time is 1.5t. Thus, the amount of resist etched away is

$$R_{resist}\left(1.5t\right) = \frac{1.5}{2.5} R_{oxide}\, t = \frac{1.5}{2.5} 140\ nm = 84\ nm$$

The absolute minimum resist thickness is 84 nm. The nominal thickness of the resist would need to be greater to account for variations in resist thickness (e.g., when coating over topography), resist etch rate (uniformity of etch in the chamber, loading effects, etc.), and the presence of horizontal etching components.

1.2. For a certain process, a 300 keV phosphorous implant is masked well by a 1.0 μm thick photoresist film.

a) For this implant process, how many multiples of the straggle does this resist thickness represent?
b) If the implant energy is increased to 450 keV, how much should the photoresist thickness be increased?
c) If the dopant is also changed to arsenic (with the energy at 450 keV), what resist thickness will be needed?

a) For phosphorous, the project range R_p and the straggle ΔR_p are given approximately by

$$R_p = 5.8E^{0.80} \quad \text{and} \quad \Delta R_p = 4.8E^{0.50}$$, with E in keV and R_p and ΔR_p in nm. Thus,

$R_p = 5.8(300)^{0.80} = 556\ nm$, $\Delta R_p = 4.8(300)^{0.50} = 83\ nm$, $1000\ nm = R_p + m\Delta R_p$, thus m = 5.34.

b) $R_p = 5.8(450)^{0.80} = 769\ nm$, $\Delta R_p = 4.8(450)^{0.50} = 105\ nm$,
$thickness = R_p + 5.34\Delta R_p = 1330\ nm$.

b) For arsenic, the project range is $R_p = 0.49E^{1.11}$. The straggle is the same as in b).

$$R_p = 0.49(450)^{1.11} = 432\ nm, \quad thickness = R_p + 5.34\Delta R_p = 992\ nm$$

1.3. A photoresist gives a final resist thickness of 320 nm when spun at 2800 rpm.
a) What spin speed should be used if a 290 nm thick coating of this same resist is desired?
b) If the maximum practical spin speed for 200 mm wafers is 4000 rpm, at what thickness would a lower viscosity formulation of the resist be required?

For resist thickness *d*, the impact of changing spin speed is given by $\frac{d_2}{d_1} = \frac{\sqrt{\omega_1}}{\sqrt{\omega_2}}$

a) The new spin speed needed would be $\omega_2 = \omega_1 \left(\frac{d_1}{d_2}\right)^2 = 2800\left(\frac{320}{290}\right)^2 = 3409\ rpm$.

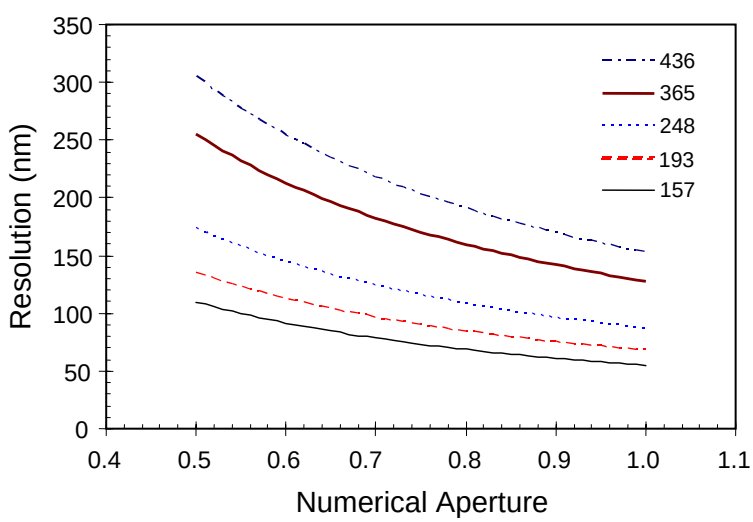
b) The minimum thickness for the same viscosity would be $d_2 = d_1 \frac{\sqrt{\omega_1}}{\sqrt{\omega_2}} = 320\left(\frac{2800}{4000}\right)^{0.5} = 268\ nm$.

For a thinner resist, a lower viscosity resist formulation would be required.

1.4. Resolution in optical lithography scales with wavelength and numerical aperture according to a modified Rayleigh criterion:

$$R = k_1 \frac{\lambda}{NA}$$

where k_1 can be thought of as a constant for a given lithographic approach and process. Assuming k_1 = 0.35, plot resolution versus numerical aperture over a range of NAs from 0.5 to 1.0 for the common lithographic wavelengths of 436 nm, 365 nm, 248 nm, 193 nm, and 157 nm. From this list, what options (NA and wavelength) are available for printing 90 nm features?



For 90nm features, the options are:

Wavelength (nm)	436	365	248	193	157
NA	N/A	N/A	0.96	0.75	0.61

(assuming immersion is not an option at 436 or 365 nm, so that the max NA is limited to 1.0).

Chapter 2

2.1. Derive equations (2.5).

Starting with equations (2.3): $\vec{\nabla}\times\vec{H}-\varepsilon\frac{\partial\vec{E}}{\partial t}=\sigma\vec{E}$, $\vec{\nabla}\times\vec{E}+\mu\frac{\partial\vec{H}}{\partial t}=0$

Taking the curl of the second equation, then substituting in the first,

$$\vec{\nabla}\times\vec{\nabla}\times\vec{E}+\vec{\nabla}\times\left(\frac{\partial\vec{B}}{\partial t}\right)=-\nabla^2\vec{E}+\mu\frac{\partial}{\partial t}\left(\vec{\nabla}\times\vec{H}\right)=-\nabla^2\vec{E}+\mu\frac{\partial}{\partial t}\left(\varepsilon\frac{\partial\vec{E}}{\partial t}+\sigma\vec{E}\right)=0$$

Rearranging,

$$\nabla^2\vec{E}=\mu\frac{\partial}{\partial t}\left(\varepsilon\frac{\partial\vec{E}}{\partial t}+\sigma\vec{E}\right)=\mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}+\mu\sigma\frac{\partial\vec{E}}{\partial t}$$

A similar procedure is used to derive the magnetic field wave equation:

$$\vec{\nabla}\times\vec{\nabla}\times\vec{H}-\varepsilon\vec{\nabla}\times\left(\frac{\partial\vec{E}}{\partial t}\right)=-\nabla^2\vec{H}-\varepsilon\frac{\partial}{\partial t}\left(\vec{\nabla}\times\vec{E}\right)=-\nabla^2\vec{H}-\varepsilon\frac{\partial}{\partial t}\left(-\mu\frac{\partial\vec{H}}{\partial t}\right)=\sigma\vec{\nabla}\times\vec{E}=-\sigma\mu\frac{\partial\vec{H}}{\partial t}$$

$$\nabla^2\vec{H}=-\varepsilon\frac{\partial}{\partial t}\left(-\mu\frac{\partial\vec{H}}{\partial t}\right)+\sigma\mu\frac{\partial\vec{H}}{\partial t}=\varepsilon\mu\frac{\partial^2\vec{H}}{\partial t^2}+\sigma\mu\frac{\partial\vec{H}}{\partial t}$$

2.2. Show that the spherical wave of equation (2.27) is in fact a solution to the Helmholtz equation.

Plug the equation for a spherical wave into the Helmholtz equation:

$$\nabla^2\vec{U}(P)+\boldsymbol{k}^2\vec{U}(P)=0,\quad U(r)=\frac{Ae^{i\boldsymbol{k}r}}{r},\quad r=\sqrt{x^2+y^2+z^2}$$

In spherical coordinates:

$$\nabla^2U=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial U}{\partial\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2U}{\partial\phi^2}=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right)$$

$$r^2\frac{\partial U}{\partial r}=\left(ikr^2-r\right)U,\quad \frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right)=\left(2ikr-1\right)U+\left(ikr^2-r\right)\left(ik-\frac{1}{r}\right)U$$

$$\nabla^2U=\frac{1}{r^2}\left(2ikr-1+-kr^2-ikr-ikr+1\right)U=-k^2U\quad\text{QED.}$$

2.3. Using equation (2.38), determine the SI units for intensity.

$$I = \left|\left\langle \vec{S} \right\rangle\right| = \frac{n}{2c\mu}|A|^2 = \frac{1}{2}\sqrt{\frac{\varepsilon}{\mu}}|A|^2$$

Using the first expression, the amplitude of the electric filed, *A*, has SI units of V/m. The speed of light has SI units of m2/s. The magnetic permeability has SI units of henries/m or V-s/(A-m). Refractive index is dimensionless. Thus,

$$\frac{(V/m)^2}{(m/s)(Vs/Am)} = \frac{VA}{m^2} = W/m^2$$

Using the second expression, dielectric constant has SI units of farads/m or C/(V-m). Thus,

$$\sqrt{\frac{C/Vm}{Vs/Am}}(V/m)^2 = \sqrt{\frac{A^2}{V^2}}(V/m)^2 = \frac{VA}{m^2} = W/m^2$$

Alternately, $Z = \sqrt{\frac{\varepsilon}{\mu}}$ = impedance, with units of Ohms. Thus, $\Omega(V/m)^2 = AV/m^2 = W/m^2$

Some students will break the units down to there lowest levels: $1W = 1\,kgm^2/s^3$ so that the units of intensity become kg/s^3.

2.4. Derive equations (2.45) for the isolated space and the dense line/space pattern. Some of the theorems from Appendix C may prove useful.

In 1-D, $E_i = 1$, $\quad T_m(f_x) = \int_{-\infty}^{\infty} t_m(x) e^{-2\pi i(f_x x)} dx$

Isolated Space: $\quad T_m(f_x) = \int_{-w/2}^{w/2} e^{-2\pi i(f_x x)} dx = \left.\frac{e^{-2\pi i(f_x x)}}{-2\pi i f_x}\right|_{-w/2}^{w/2} = \frac{e^{-i\pi f_x w} - e^{i\pi f_x w}}{-2\pi i f_x}$

$$T_m(f_x) = \frac{-2i\sin(\pi f_x w)}{-2\pi i f_x} = \frac{\sin(\pi f_x w)}{\pi f_x}$$

Dense line/space: $\quad T_m(f_x) = \sum_{j=-\infty}^{\infty} \int_{jP-w/2}^{jP+w/2} e^{-2\pi i(f_x x)} dx$

$$= \sum_{j=-\infty}^{\infty} \frac{e^{-i2\pi f_x(jP+w/2)} - e^{-i2\pi f_x(jP-w/2)}}{-2\pi i f_x} = \sum_{j=-\infty}^{\infty} \frac{\sin(\pi f_x w)}{\pi f_x} e^{-i2\pi f_x jP}$$

Using identity the Dirac Comb identity from Appendix C,

$$T_m(f_x) = \frac{1}{p}\sum_{j=-\infty}^{\infty} \frac{\sin(\pi f_x w)}{\pi f_x}\delta\left(f_x - \frac{j}{p}\right). \quad \text{QED.}$$

Alternatively, the dense line/space pattern transmittance can be described as $isospace * \sum_{j=-\infty}^{\infty}\delta(x - jp)$.

Thus, $T_m(f_x) = \mathcal{F}\{isospace\}\mathcal{F}\left\{\sum_{j=-\infty}^{\infty}\delta(x - jp)\right\}$. Applying a transform pair from Table 2.1 for the Comb function plus the scaling property of the Fourier transform gives the final answer.

2.5. Show that the diffraction pattern for an dense line/space pattern becomes the diffraction pattern for an isolated space as the pitch becomes infinite (for a constant spacewidth w). This is equivalent to proving that

$$\lim_{p\to\infty}\frac{1}{p}\sum_{j=-\infty}^{\infty}\delta\left(f_x - \frac{j}{p}\right) = 1$$

Let $\Delta = 1/p$. Thus, we want to show that

$$\lim_{\Delta\to 0}\sum_{j=-\infty}^{\infty}\delta(f_x - j\Delta)\Delta = 1$$

But the left hand side, in the limit, is just an integral:

$$\lim_{\Delta\to 0}\sum_{j=-\infty}^{\infty}\delta(f_x - j\Delta)\Delta = \int_{-\infty}^{\infty}\delta(f_x - x)dx = 1. \quad \text{QED.}$$

2.6. Complimentary mask features (for example, an isolated line and an isolated space of the same width) are defined by

$$t_m^c(x,y) = 1 - t_m(x,y)$$

Prove that the diffraction patterns of complimentary mask features are given by

$$T_m^c(f_x, f_y) = \delta(f_x, f_y) - T_m(f_x, f_y)$$

Use this expression to derive the diffraction pattern of an isolated line.

Taking the Fourier transform of the first equation:

$$\mathcal{F}\{t_m^c(x,y)\} = \mathcal{F}\{1\} - \mathcal{F}\{t_m(x,y)\} \Rightarrow T_m^c(f_x, f_y) = \delta(f_x, f_y) - T_m(f_x, f_y)$$

For a 1-D space, $T_m(f_x) = \dfrac{\sin(\pi w f_x)}{\pi f_x}$. Thus, the complimentary line has a diffraction pattern

$$T_m^c(f_x, f_y) = \delta(f_x, f_y) - \frac{\sin(\pi w f_x)}{\pi f_x}$$

2.7. Show that the Fourier transform is a linear operation, that is, show that for two functions $f(x,y)$ and $g(x,y)$, and two constants a and b,

$$\mathcal{F}\{af(x,y) + bg(x,y)\} = aF(f_x, f_y) + bG(f_x, f_y)$$

Applying the definition of the Fourier transform,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \big(af(x,y) + bg(x,y)\big) e^{-2\pi i(f_x x + f_y y)} dx\,dy$$

$$= a\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y) e^{-2\pi i(f_x x + f_y y)} dx\,dy + b\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) e^{-2\pi i(f_x x + f_y y)} dx\,dy$$

$$= aF(f_x, f_y) + bG(f_x, f_y)$$

2.8. Prove the shift theorem of the Fourier transform:

$$\text{If } \mathcal{F}\{g(x,y)\} = G(f_x, f_y), \quad \mathcal{F}\{g(x-a, y-b)\} = G(f_x, f_y) e^{-i2\pi(f_x a + f_y b)}$$

Applying the definition of the Fourier transform, then letting $x' = x - a$ and $y' = y - b$,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x-a, y-b) e^{-2\pi i(f_x x + f_y y)} dx\,dy = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x', y') e^{-2\pi i\left(f_x[x'+a] + f_y[y'+b]\right)} dx'\,dy'$$

$$= e^{-2\pi i\left(f_x a + f_y b\right)} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x', y') e^{-2\pi i\left(f_x x' + f_y y'\right)} dx'\,dy' = G(f_x, f_y) e^{-i2\pi(f_x a + f_y b)}$$

2.9. Prove the similarity theorem of the Fourier transform:

$$\text{If } \mathcal{F}\{g(x,y)\} = G(f_x, f_y), \quad \mathcal{F}\{g(ax, by)\} = \frac{1}{|ab|} G\left(\frac{f_x}{a}, \frac{f_y}{b}\right)$$

Applying the definition of the Fourier transform, then letting $x' = ax$ and $y' = by$,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(ax, by) e^{-2\pi i(f_x x + f_y y)} dx\,dy = \frac{1}{|ab|}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x', y') e^{-2\pi i\left(f_x[x'/a] + f_y[y'/b]\right)} dx'\,dy'$$

$$= \frac{1}{|ab|}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x', y') e^{-2\pi i\left(\frac{f_x}{a}x' + \frac{f_y}{b}y'\right)} dx'\,dy'$$

Why the absolute value sign out in front (the factor $1/|ab|$)? Suppose a is negative. Then the limits of the integration of x' would have to go from $+\infty$ to $-\infty$. Multiplying the integral by -1 would put the range of integration back to its original $-\infty$ to $+\infty$. Thus, the multiplier in front of the integral would be $-1/a$ if a is negative, and $+1/a$ if a is positive. This is equivalent to $1/|a|$. Of course, the same argument holds for b.

2.10. Prove the convolution theorem of the Fourier transform:

$$\text{If } \mathcal{F}\{g(x,y)\} = G(f_x, f_y) \text{ and } \mathcal{F}\{h(x,y)\} = H(f_x, f_y),$$

$$\mathcal{F}\left\{\int\int_{-\infty}^{\infty} g(\xi,\eta)h(x-\xi,y-\eta)d\xi\, d\eta\right\} = G(f_x, f_y)H(f_x, f_y)$$

Applying the definition of the Fourier transform,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(\int\int_{-\infty}^{\infty} g(\xi,\eta)h(x-\xi,y-\eta)d\xi\, d\eta\right)e^{-2\pi i(f_x x + f_y y)}dx\, dy$$

Changing the order of integration,

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(\xi,\eta)\left(\int\int_{-\infty}^{\infty} h(x-\xi,y-\eta)e^{-2\pi i(f_x x + f_y y)}dx\, dy\right)d\xi\, d\eta$$

Applying the shift theorem to the inner integral,

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(\xi,\eta)\left(H(f_x, f_y)e^{-i2\pi(f_x\xi + f_y\eta)}\right)d\xi\, d\eta = H(f_x, f_y)\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(\xi,\eta)e^{-i2\pi(f_x\xi + f_y\eta)}d\xi\, d\eta$$
$$= H(f_x, f_y)G(f_x, f_y)$$

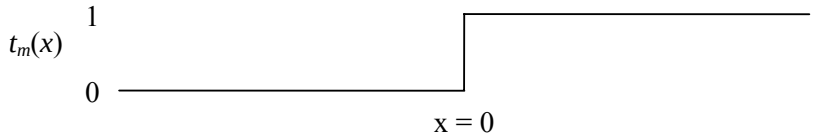
2.11. Prove the scaling property of the Dirac delta function [in Appendix C, equation C.10].

Prove that: $\delta(ax) = \dfrac{\delta(x)}{|a|}$.

The definition of a delta function is any function such that $\delta(x) = 0$ when $x \neq 0$ and $\int_{-\infty}^{\infty}\delta(x)\,dx = 1$.

Now, integrate the left hand side of the above equation to be proved:
$\int_{-\infty}^{\infty}\delta(ax)\,dx = ?$. Letting $x' = ax$, $\int_{-\infty}^{\infty}\delta(ax)\,dx = \dfrac{1}{|a|}\int_{-\infty}^{\infty}\delta(x')\,dx' = \dfrac{1}{|a|}$. Thus, the quantity $|a|\delta(ax)$ satisfies the definition of a Dirac delta function. QED. (If you are wondering where the absolute value around a came from, see the solution to problem 2.9.)

2.12. Derive the diffraction pattern of an isolated edge:



Straightforward integration poses a problem:

$$T_m(f_x) = \int_0^\infty e^{-2\pi i f_x x} dx = \left. \frac{e^{-2\pi i f_x x}}{-2\pi i f_x} \right|_0^\infty = \frac{i}{2\pi f_x} \left[\lim_{x\to\infty} \left(e^{-2\pi i f_x x} \right) - 1 \right]$$

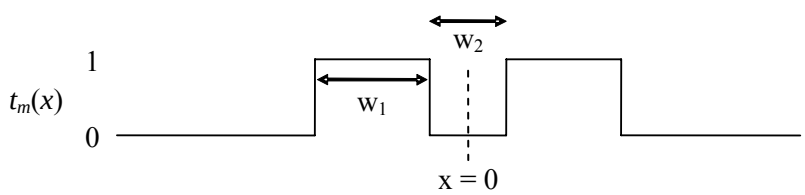
What is the value of this limit? Because the complex exponential oscillates, the limit is not obvious. There are several possible approaches to solving this problem, but the most rigorous involves using the Cauchy principle value of the integral. I'll skip that approach as too detailed, and instead use the general form of the Fourier transform of an integral:

$$\mathcal{F}\left\{ \int_{-\infty}^{x} g(t)dt \right\} = \frac{G(f_x)}{i2\pi f_x} + \frac{G(0)}{2}\delta(f_x)$$

Letting $g(t) = \delta(t)$, the integral defines a step function, as described in Appendix C. And since the Fourier transform of the delta function is just 1,

$$\mathcal{F}\left\{ \int_{-\infty}^{x} \delta(t)dt \right\} = \mathcal{F}\{u(x)\} = \frac{1}{i2\pi f_x} + \frac{1}{2}\delta(f_x)$$

2.13. Derive the diffraction pattern of an isolated double space pattern:



The diffraction pattern for one space of width w_1, centered at $x = 0$:

$$T_m(f_x) = \frac{\sin(\pi f_x w_1)}{\pi f_x}$$

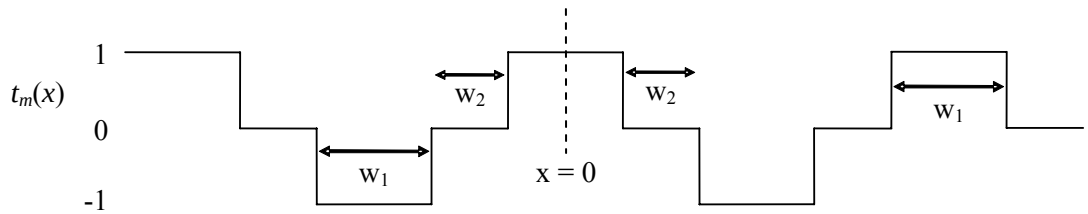
Use the shift theorem to shift that space to $+(w_1+w_2)/2$:

$$T_m(f_x) = \frac{\sin(\pi f_x w_1)}{\pi f_x} e^{-i\pi f_x (w_1 + w_2)}$$

Use superposition to add the second space shifted by $-(w_1+w_2)/2$:

$$T_m(f_x) = \frac{\sin(\pi f_x w_1)}{\pi f_x} e^{-i\pi f_x (w_1+w_2)} + \frac{\sin(\pi f_x w_1)}{\pi f_x} e^{i\pi f_x (w_1+w_2)} = 2\cos(\pi f_x (w_1 + w_2)) \frac{\sin(\pi f_x w_1)}{\pi f_x}$$

2.14. Derive the diffraction pattern of an alternating phase shift mask (a pattern of lines and spaces where every other space is shifted in phase by 180°, resulting in a transmittance of −1):



The diffraction pattern for a repeating pattern of spaces of width w_1 and pitch $2p$, where $p = w_1+w_2$, centered at $x = 0$:

$$T_m(f_x) = \frac{1}{2p} \sum_{j=-\infty}^{\infty} \frac{\sin(\pi f_x w_1)}{\pi f_x} \delta\left(f_x - \frac{j}{2p} \right)$$

The diffraction pattern of a second repeating pattern of spaces of width w_1 and pitch $2p$, shifted by p and with transmittance -1:

$$T_m(f_x) = -\frac{1}{2p} \sum_{j=-\infty}^{\infty} \frac{\sin(\pi f_x w_1)}{\pi f_x} \delta\left(f_x - \frac{j}{2p} \right) e^{-i2\pi f_x p}$$

Using superposition to add these two arrays together:

$$T_m(f_x) = \frac{1}{2p} \sum_{j=-\infty}^{\infty} \frac{\sin(\pi f_x w_1)}{\pi f_x} \delta\left(f_x - \frac{j}{2p} \right) \left(1 - e^{-i2\pi f_x p} \right)$$

But, since $f_x = j/2p$ are the only non-zero values of the diffraction pattern, we can substitute this value everywhere for f_x:

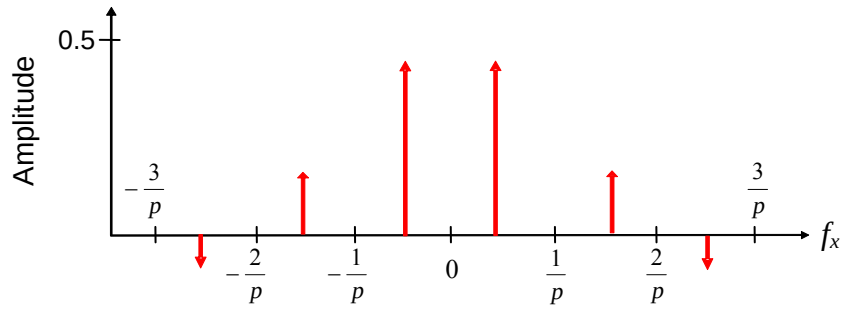
$$T_m(f_x) = \sum_{j=-\infty}^{\infty} \left(1 - e^{-i(j\pi)} \right) \frac{\sin(j\pi w_1 / 2p)}{j\pi} \delta\left(f_x - \frac{j}{2p} \right) = \sum_{j=-\infty}^{\infty} \left(1 - (-1)^j \right) \frac{\sin(j\pi w_1 / 2p)}{j\pi} \delta\left(f_x - \frac{j}{2p} \right)$$

Notice that all the even values of j disappear.

Graph the resulting diffraction pattern out to the $\pm 5^{th}$ diffraction orders for the case of $w_1 = w_2$.

The diffraction pattern will have diffraction orders spaced out at intervals of $1/2p$ with amplitudes

$$a_j = \begin{cases} 0, & j = even \\ 2\dfrac{\sin(j\pi/4)}{j\pi}, & j = odd \end{cases}, \quad a_1 = a_{-1} = \frac{\sqrt{2}}{\pi}, \quad a_3 = a_{-3} = \frac{\sqrt{2}}{3\pi}, \quad a_5 = a_{-5} = -\frac{\sqrt{2}}{5\pi}, \quad \text{etc.}$$



2.15. Prove equation (2.59).

Prove that $\mathcal{F}\{\mathcal{F}\{g(x,y)\}\} = g(-x,-y)$. Applying the definition of the Fourier transform:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)e^{-2\pi i(f_x x + f_y y)}dx\,dy\right)e^{-2\pi i(f_x x' + f_y y')}df_x\,df_y$$

Now change the order of integration:

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)\left(\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-2\pi i\left(f_x(x+x') + f_y(y+y')\right)}df_x\,df_y\right)dx\,dy$$

But the inner integral is just a delta function shifted to the position (-x',-y').

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)\delta(x+x', y+y')dx\,dy = g(-x',-y'). \quad \text{QED.}$$

Alternately, proving this identity is equivalent to proving that $\mathcal{F}\{g(x,y)\} = \mathcal{F}^{-1}\{g(-x,-y)\}$. Applying the definition of the inverse Fourier transform,

$$\mathcal{F}^{-1}\{g(-x,-y)\} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(-x,-y)e^{2\pi i(f_x x + f_y y)}dx\,dy$$

Let a = -x, b = -y,

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(a,b)e^{-2\pi i(f_x a + f_y b)}da\,db$$

But this is just the definition of the Fourier transform. QED.

2.16. In section 2.3, it was claimed that illuminating a mask with a plane wave at an angle θ' resulted, under the Kirchhoff boundary condition assumption, in a diffraction pattern equal to the normally incident diffraction pattern shifted by an amount $\sin\theta'/\lambda$. Prove this statement using the definition of the Fraunhofer diffraction pattern, equation (2.44).

$$T_m(f_x, f_y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} E_i(x,y)t_m(x,y)e^{-2\pi i(f_x x + f_y y)}dx\,dy$$

For illumination tilted in one direction only, use $E_i(x,y) = e^{i2\pi \sin\theta' x/\lambda}$.

$$T_m(f_x, f_y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{i2\pi \sin\theta' x/\lambda} t_m(x,y)e^{-2\pi i(f_x x + f_y y)}dx\,dy$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} t_m(x,y)e^{-2\pi i\left((f_x - \sin\theta'/\lambda)x + f_y y\right)}dx\,dy = T_m(f_x - \sin\theta'/\lambda, f_y) = T_m(f_x - f_x', f_y)$$

Of course, the same is true for a tilt in the y-direction. This is nothing more than a proof of the shift theorem of the Fourier transform.

2.17. Consider a binary pattern of lines and spaces. Which diffraction orders pass through the lens under these circumstances:

(a) λ = 248 nm, NA = 0.8, pitch = 300 nm, on-axis coherent illumination
(b) Same as (a), but pitch = 400 nm
(c) Same as (b), but illumination tilted by an angle sinθ' = 0.5

a) Spatial frequency = j/p. Cutoff frequency = NA/λ. Max $j \le pNA/\lambda = 0.96$. Thus, only the zero order makes it through the lens.
b) Max $j \le pNA/\lambda = 1.29$. Thus, the zero and $\pm 1^{st}$ orders make it through the lens.
c) Spatial frequency = $j/p + 0.5/\lambda$. Cutoff frequency = NA/λ. Max $j \le p(NA\text{-}0.5)/\lambda = 0.48$. Thus, the max order is 0. Min $j \ge -p(NA+0.5)/\lambda = -2.096$. Thus, the min order is –2. The 0, –1 and –2 orders go through the lens.

2.18. Equations (2.65) and (2.67) give the electric field and intensity images for three-beam images, where the 0 and $\pm 1^{st}$ diffraction orders are used to form the image. Derive similar expressions for two-beam images, where only the 0 and $+1^{st}$ diffraction orders are captured by the lens.

For two beam imaging,

$$E(x) = \sum_{j=0}^{1} \frac{\sin(j\pi w/p)}{j\pi} e^{i2\pi jx/p} = \frac{w}{p} + \frac{\sin(\pi w/p)}{\pi} e^{i2\pi x/p} = a_0 + a_1 e^{i2\pi x/p}$$

The intensity becomes

$$I(x) = E(x)E^*(x) = \left(a_0 + a_1 e^{i2\pi x/p}\right)\left(a_0 + a_1 e^{-i2\pi x/p}\right) = a_0^2 + a_1^2 + a_0 a_1\left(e^{i2\pi x/p} + e^{-i2\pi x/p}\right)$$

$$I(x) = a_0^2 + a_1^2 + 2a_0 a_1 \cos\left(2\pi x/p\right)$$

2.19. For a repeating line/space pattern and coherent illumination, derive expressions for the aerial image intensity at the center of the line and the center of the space as a function of the number of diffraction orders captured.

For lines and spaces and coherent illumination, $E(x) = a_0 + 2\sum_{j=1}^{N} a_j \cos(2\pi jx/p)$. At the center of the space, $x = 0$. Thus,

$E(x=0) = \sum_{j=-N}^{N} a_j$, $\quad I(x=0) = \left(\sum_{j=-N}^{N} a_j\right)\left(\sum_{j=-N}^{N} a^*_j\right)$. For the common case where the diffraction order amplitudes are real,

$$I(x=0) = \left(\sum_{j=-N}^{N} a_j\right)^2$$

At the center of the line, $x = p/2$.

$$E(x) = a_0 + 2\sum_{j=1}^{N} a_j \cos(j\pi) = \sum_{j=-N}^{N} a_j (-1)^j$$

2.20. For λ = 248 nm, NA = 0.8, and pitch = 400 nm, below what value of σ is the image entirely made up of three-beam interference? At what σ value does one-beam imaging first appear?

From Table 2.3, imaging is entirely three-beam when $\frac{\lambda}{pNA} \le 1 - \sigma$. Thus, $\sigma \le 1 - \frac{\lambda}{pNA}$ = 0.225.

From Table 2.3, imaging begins to have a one-beam component when $\sqrt{1-\sigma^2} \le \frac{\lambda}{pNA}$. Thus, $\sigma \ge \sqrt{1 - \left(\frac{\lambda}{pNA}\right)^2}$ = 0.632.

2.21. For coherent illumination and a lens of a given NA and λ, derive a formula for the electric field image of an isolated edge. The result will be in terms of the sine integral, defined as

$$Si(z) = \int_0^z \frac{\sin y}{y} dy$$

$$E(x) = \int_{-NA/\lambda}^{NA/\lambda} \left(\frac{1}{2}\delta(f_x) - \frac{i}{2\pi f_x} \right) e^{i2\pi f_x x} df_x = \frac{1}{2} - \frac{i}{2\pi} \int_{-NA/\lambda}^{NA/\lambda} \frac{e^{i2\pi f_x x}}{f_x} df_x$$

Breaking up the complex exponential using Euler's relation,

$$E(x) = \frac{1}{2} - \frac{i}{2\pi} \int_{-NA/\lambda}^{NA/\lambda} \frac{\cos(2\pi f_x x)}{f_x} df_x + \frac{1}{2\pi} \int_{-NA/\lambda}^{NA/\lambda} \frac{\sin(2\pi f_x x)}{f_x} df_x$$

The middle term will vanish since the integrand is odd. Since the last term has an even integrand,

$$E(x) = \frac{1}{2} + \frac{1}{\pi} \int_{0}^{NA/\lambda} \frac{\sin(2\pi f_x x)}{f_x} df_x = \frac{1}{2} + \frac{1}{\pi} \int_{0}^{2\pi NAx/\lambda} \frac{\sin(y)}{y} dy = \frac{1}{2} + \frac{1}{\pi} Si\left(\frac{2\pi NAx}{\lambda} \right)$$

Chapter 3

3.1. Derive equation (3.5).

Derive $placement\ error = \dfrac{Z_1 \lambda}{NA}$.

The first Zernike polynomial term gives $W(R,\phi) = Z_1 R \cos\phi$. For y-oriented lines, the diffraction pattern will be along the x-axis so that $\phi = 0$. Converting R to spatial frequency,

$W(R,\phi) = Z_1 f_x \lambda / NA$ and $P(f_x, f_y) = P_{ideal}(f_x, f_y) e^{i2\pi W(R,\phi)} = P_{ideal}(f_x, f_y) e^{i2\pi Z_1 f_x \lambda / NA}$

The electric field image without aberrations will be will be

$E_{ideal}(x,y) = \mathcal{F}^{-1}\{T_m(f_x, f_y) P_{ideal}(f_x, f_y)\}$

With aberrations,

$E(x,y) = \mathcal{F}^{-1}\{T_m(f_x, f_y) P(f_x, f_y)\} = \mathcal{F}^{-1}\{T_m(f_x, f_y) P_{ideal}(f_x, f_y) e^{i2\pi Z_1 f_x \lambda / NA}\}$

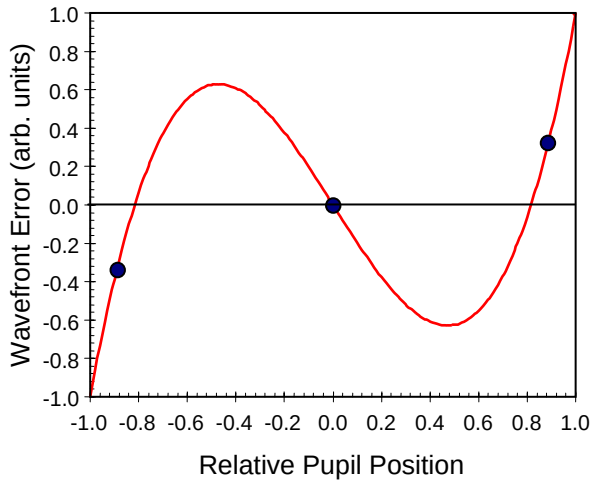
Applying the shift theorem to the inverse Fourier transform,

$E(x,y) = E_{ideal}(x + Z_1 \lambda / NA, y)$. Thus, the pattern placement error is $-Z_1 \lambda / NA$.

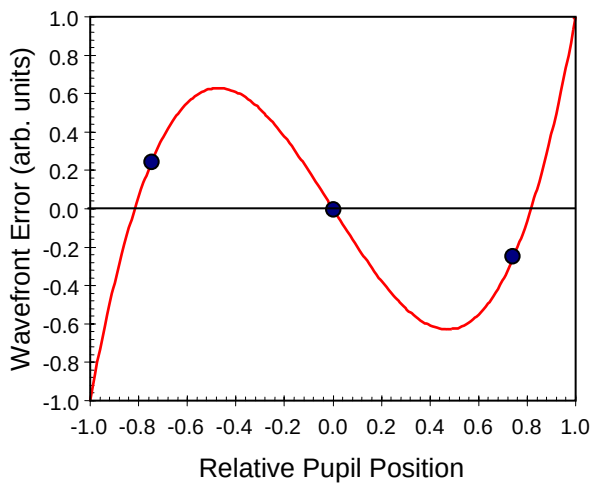
3.2. Equation (3.7), which was derived for coherent illumination, predicts that coma will result in a negative placement error for lines and spaces when the pitch is less than $\sqrt{1.5}\lambda / NA$, a positive placement error for pitches greater than this amount, and no placement error when the pitch equals $\sqrt{1.5}\lambda / NA$. Show why this is true using pictures of the wavefront error and the diffraction pattern.

Pitch = $\sqrt{1.5}\lambda / NA$ corresponds to a relative pupil position of $1/\sqrt{1.5} \approx 0.816$

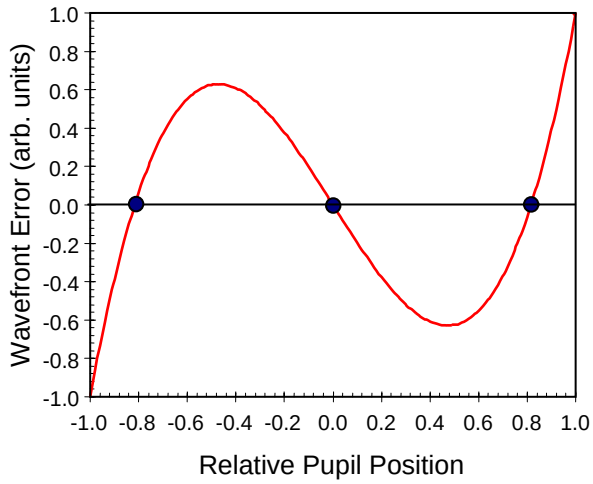
a) Pitch less than $\sqrt{1.5}\lambda / NA$ means the first order will be at R > 0.816. This gives a positive tilt, which then gives a negative placement error.



b) Pitch greater than $\sqrt{1.5}\lambda / NA$ means the first order will be at R < 0.816. This gives a negative tilt, which then gives a positive placement error.



c) Pitch equal to $\sqrt{1.5}\lambda / NA$ means the first order will be at R = 0.816. This gives no tilt and no placement error.



3.3. Using the Zernike polynomial term F_4, explain the effect of 3rd-order astigmatism on *x*-oriented lines and spaces versus *y*-oriented lines and spaces.

3rd order astigmatism gives $W(R,\phi) = Z_4 R^2 \cos 2\phi$.

For *y*-oriented lines, $\phi = 0$ and $W(R,\phi) = Z_4 R^2$. Thus, astigmatism acts like defocus.

For *x*-oriented lines, $\phi = 90°$ and $W(R,\phi) = -Z_4 R^2$. Thus, astigmatism acts like defocus, but in the opposite direction as for the y-oriented lines.

3.4. Calculate the RMS wavefront error, W_{RMS}, and the resulting Strehl ratio for the case when

a) Only 3rd-order spherical aberration is present

b) Only 3rd-order x-coma is present

Note that $$W_{RMS}^2 = \frac{1}{\pi}\int_0^1\int_0^{2\pi} W^2(R,\phi)\,R\,dR\,d\phi$$

a) $$W_{RMS}^2 = \frac{1}{\pi}\int_0^1\int_0^{2\pi}\left(Z_8\left[6R^4 - 6R^2 + 1\right]\right)^2 R\,dR\,d\phi = 2Z_8{}^2\int_0^1\left(6R^4 - 6R^2 + 1\right)^2 R\,dR$$

$$= 2Z_8{}^2\int_0^1\left(36R^8 + 36R^4 - 72R^6 + 12R^4 - 12R^2 + 1\right)R\,dR$$

$$= 2Z_8{}^2\int_0^1\left(36R^9 - 72R^7 + 48R^5 - 12R^3 + R\right)dR$$

$$= 2Z_8{}^2\left(\frac{36R^{10}}{10} - \frac{72R^8}{8} + \frac{48R^6}{6} - \frac{12R^4}{4} + \frac{R^2}{2}\right)\Bigg|_0^1 = 0.2Z_8{}^2$$

$$W_{RMS} = \frac{|Z_8|}{\sqrt{5}}$$

b) $W_{RMS}^2 = \frac{1}{\pi}\int_0^1\int_0^{2\pi}\left(Z_6\left[3R^2-2\right]R\cos\phi\right)^2 RdRd\phi = \frac{Z_6^{\,2}}{\pi}\int_0^1\left(3R^2-2\right)^2 R^3dR\int_0^{2\pi}\cos^2\phi d\phi$

$$= \frac{Z_6^{\,2}}{\pi}\int_0^1\left(9R^4-12R^2+4\right)R^3dR\left[\frac{1}{2}\left(\phi+\cos\phi\sin\phi\right)\right]_0^{2\pi} = Z_6^{\,2}\int_0^1\left(9R^7-12R^5+4R^3\right)dR$$

$$= Z_6^{\,2}\left(\frac{9R^8}{8}-\frac{12R^6}{6}+\frac{4R^4}{4}\right)\Bigg|_0^1 = \frac{Z_6^{\,2}}{8}$$

$$W_{RMS} = \frac{\left|Z_6\right|}{\sqrt{8}}$$

3.5. Consider coherent imaging of equal lines and spaces where only the zero and ±first diffraction orders enter the lens. Derive an expression, similar to equation (3.24), that shows how spherical aberration results in a pitch-dependent paraxial focus error.

For spherical, $W(R,\phi) = Z_8\left[6R^4-6R^2+1\right]$. Thus, the difference between the zero and first orders is $\Delta W = Z_8\left[6R^4-6R^2\right]$ where $R = \frac{\lambda}{pNA}$. For defocus, the wavefront difference between zero and first orders, in the paraxial approximation, is $\Delta W = \frac{1}{2}\delta\lambda/p^2$. Equating the two,

$$\frac{1}{2}\delta\lambda/p^2 = Z_8\left[6R^4-6R^2\right] = 6Z_8\left(\frac{\lambda}{pNA}\right)^2\left[\left(\frac{\lambda}{pNA}\right)^2-1\right]$$

$$\delta = 12Z_8\left(\frac{\lambda}{NA^2}\right)\left[\left(\frac{\lambda}{pNA}\right)^2-1\right] \quad \text{or} \quad Z_8 = \frac{\delta NA^2}{12\lambda\left[\left(\frac{\lambda}{pNA}\right)^2-1\right]}$$

3.6. Calculate the impact of long-range flare on the image log-slope.

$$I(x) = SF\cdot CF + (1-SF)I_0(x), \quad \frac{dI}{dx} = (1-SF)\frac{dI_0}{dx}$$

$$\frac{d\ln I}{dx} = \frac{d\ln I_0}{dx}\left[\frac{(1-SF)I_0(x_{edge})}{SF\cdot CF+(1-SF)I_0(x_{edge})}\right] = \frac{d\ln I_0}{dx}\left[\frac{1}{1+a}\right] \quad \text{where} \quad a = \frac{SF\cdot CF}{(1-SF)I_0(x_{edge})}.$$

3.7. Consider the case of dense equal lines and spaces (only the 0 and ±1st orders are used) imaged with coherent illumination. Show that the peak intensity of the image in the middle of the space falls off approximately quadratically with defocus for small amounts of defocus.

Assuming TE illumination, from Table 3.2,

$$I(x) = a_0^2 + 2a_1{}^2 + 4a_0a_1\cos(\Delta\Phi)\cos(2\pi x/p) + 2a_1^2\cos(4\pi x/p)$$

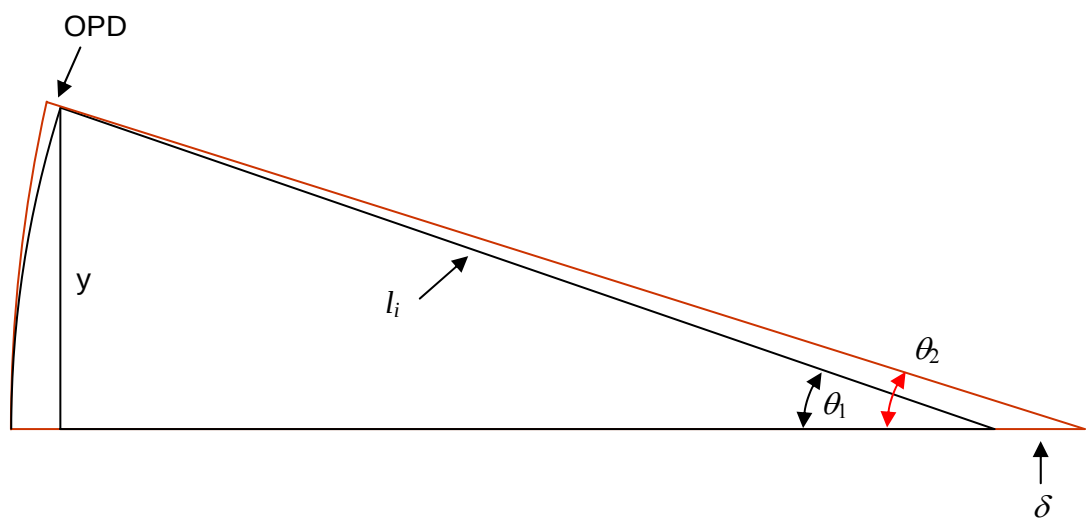
At the center of the space, $x = 0$ and $I(0) = a_0^2 + 4a_1{}^2 + 4a_0a_1\cos(\Delta\Phi)$. Using a Taylor series expansion of the cosine,

$$I(0) \approx a_0^2 + 4a_1{}^2 + 4a_0a_1\left(1 - \frac{\Delta\Phi^2}{2}\right) = I_{in-focus}(0)\left(1 - \frac{\Delta\Phi^2}{2}\right)$$

Since $\Delta\Phi$ is directly proportional to defocus, one can see that the peak intensity will fall off approximately as the defocus squared.

3.8. Using the geometry of Figure 3.18b, derive equation (3.20).

The first wavefront has radius of curvature l_i, but the defocused wavefront has radius $l_i + \delta$. Consider the geometry below (not drawn to scale to make it easier to see):



Consider the two right triangles that share the height y. The smaller triangle makes the angle θ_1 with the horizontal and has hypotenuse l_i. The larger triangle makes the angle θ_2 and has hypotenuse $l_i + \delta - \text{OPD}$. Comparing their heights and their bases gives

$$y = l_i\sin\theta_1 = (l_i + \delta - OPD)\sin\theta_2$$
$$x = l_i\cos\theta_1 = (l_i + \delta - OPD)\cos\theta_2 - \delta$$

The easiest way to solve these two equations simultaneously is to square them and then add them together, giving

$$l_i{}^2 = (l_i + \delta - OPD)^2 + \delta^2 + 2\delta(l_i + \delta - OPD)\cos\theta_2$$
$$l_i{}^2 = l_i{}^2 + (\delta - OPD)^2 + 2l_i(\delta - OPD) + \delta^2 + 2\delta l_i\cos\theta_2 + 2\delta(\delta - OPD)\cos\theta_2$$

Canceling the $l_i{}^2$ terms,

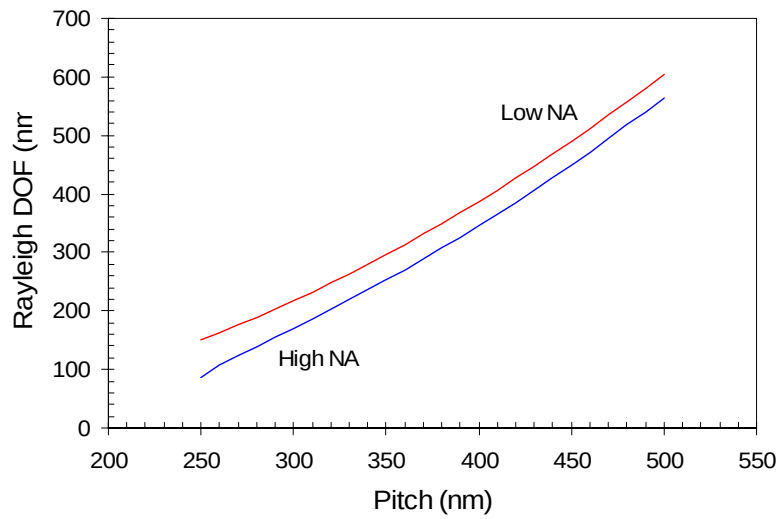
$$0 = 2l_i(\delta - OPD + \delta\cos\theta_2) + (\delta - OPD)^2 + \delta^2 + 2(\delta - OPD)\cos\theta_2$$

For l_i » δ and l_i » OPD, all of the terms in the above equation that don't have l_i can be neglected. Also, in this limit $\theta_1 \approx \theta_2$. Thus,

$0 = 2l_i(\delta - OPD + \delta\cos\theta)$ and so $0 = \delta - OPD + \delta\cos\theta$ giving $OPD = \delta(1+\cos\theta)$. What I have called OPD here is really the physical path length. To convert to OPD, simply multiply by the refractive index of the media.

3.9. Compare the depth of focus predictions of the high-NA version of the Rayleigh equation (3.53) to the paraxial version of equation (3.54) by plotting predicted DOF versus pitch (use $k_2 = 0.6$, $\lambda = 248$ nm, pitch in the range from 250 to 500 nm, and assume imaging in air).

$$DOF_{highNA} = \frac{k_2}{2}\frac{\lambda}{n(1-\cos\theta)}, \quad DOF_{lowNA} \approx k_2\frac{\lambda}{\sin^2\theta} = k_2\frac{p^2}{\lambda}$$



3.10. Consider two unit-amplitude plane waves traveling in air at angles $\pm\theta$ with respect to the z-axis (as in Figure 3.26). Show that the resulting interference patterns, depending on the polarizations of the two beams, will be

TE-polarization: $I(x) = 2 + 2\cos(4\pi x\sin(\theta)/\lambda)$
TM-polarization: $I(x) = 2 + 2\cos(2\theta)\cos(4\pi x\sin(\theta)/\lambda)$

What is the pitch of the resulting line/space image?

TE: $E_y(x) = e^{i2\pi x\sin\theta/\lambda} + e^{-i2\pi x\sin\theta/\lambda} = 2\cos(2\pi x\sin\theta/\lambda)$, $E_x = 0$, $E_z = 0$

$I(x) = |E_y(x)|^2 = 4\cos^2(2\pi x\sin\theta/\lambda) = 2 + 2\cos(4\pi x\sin\theta/\lambda)$

TM: $E_x(x) = \cos(\theta)e^{i2\pi x\sin\theta/\lambda} + \cos(\theta)e^{-i2\pi x\sin\theta/\lambda} = 2\cos(\theta)\cos(2\pi x\sin\theta/\lambda)$, $E_y = 0$

$E_z(x) = \sin(\theta)e^{i2\pi x\sin\theta/\lambda} - \sin(\theta)e^{-i2\pi x\sin\theta/\lambda} = 2i\sin(\theta)\sin(2\pi x\sin\theta/\lambda)$

$$I(x) = \left|E_x(x)\right|^2 + \left|E_z(x)\right|^2 = 4\cos^2\theta\cos^2\left(2\pi x\sin\theta/\lambda\right) + 4\sin^2\theta\sin^2\left(2\pi x\sin\theta/\lambda\right)$$
$$I(x) = 4\cos^2\theta\cos^2\left(2\pi x\sin\theta/\lambda\right) + 4\sin^2\theta\sin^2\left(2\pi x\sin\theta/\lambda\right)$$

Some trigonometric manipulations will show that $2(\cos a\cos b)^2 + 2(\sin a\sin b)^2 = 1 + \cos 2a\cos 2b$. Thus, $I(x) = 2 + 2\cos 2\theta\cos\left(4\pi x\sin\theta/\lambda\right)$.

3.11. Derive equation (3.76). What is the limit of this equation for $p \gg \lambda$?

Derive $\dfrac{DOF(immersion)}{DOF(dry)} = \dfrac{1-\sqrt{1-(\lambda/p)^2}}{n-\sqrt{n^2-(\lambda/p)^2}}$.

$$DOF = \frac{k_2}{2}\frac{\lambda}{n(1-\cos\theta)}, \quad n\sin\theta = \frac{\lambda}{p}, \quad DOF = \frac{k_2}{2}\frac{\lambda}{n\left(1-\sqrt{1-\sin^2\theta}\right)} = \frac{k_2}{2}\frac{\lambda}{n-\sqrt{n^2-n^2\sin^2\theta}}$$

$$DOF = \frac{k_2}{2}\frac{\lambda}{n-\sqrt{n^2-(\lambda/p)^2}}$$

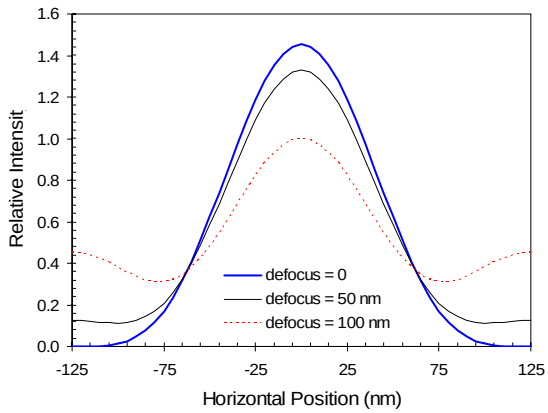
For dry lithography, $n = 1$ and $\dfrac{DOF(immersion)}{DOF(dry)} = \dfrac{1-\sqrt{1-(\lambda/p)^2}}{n-\sqrt{n^2-(\lambda/p)^2}} = \left(\dfrac{1}{n}\right)\dfrac{1-\sqrt{1-(\lambda/p)^2}}{1-\sqrt{1-(\lambda/np)^2}}$

For $p \gg \lambda$, take a Taylor series expansion of each square root: $\sqrt{1-x} \approx 1 - x/2$.

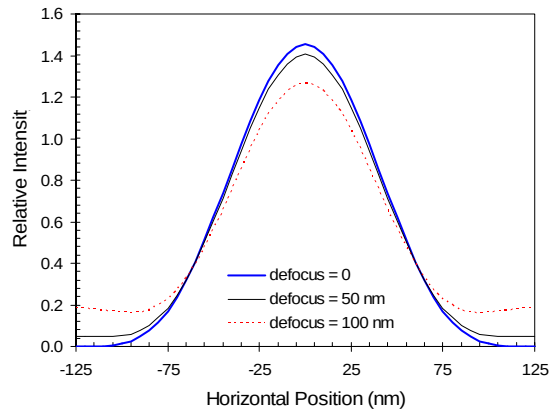
$\dfrac{DOF(immersion)}{DOF(dry)} \approx \left(\dfrac{1}{n}\right)\dfrac{\frac{1}{2}(\lambda/p)^2}{\frac{1}{2}(\lambda/np)^2} = n$. Thus, the minimum improvement in DOF using immersion is equal to the refractive index of the fluid.

3.12. Consider a coherent, three-beam image of a line/space pattern under TE illumination ($NA = 0.93$, $\lambda = 193$ nm, linewidth = 100 nm, pitch = 250 nm). Plot the resulting aerial image at best focus and at defocus values of 50 and 100 nm for:

a) imaging in air
b) immersion imaging ($n_{fluid} = 1.44$)



(a)



(b)

3.13. Consider an in-focus, TE image where only the zero order and one of the first orders pass through the lens. For this image, what are the values of β_0, β_1, and β_2 in equation (3.86)? What is the value of the NILS for the case of equal lines and spaces?

$$I(x) = a_0^2 + a_1{}^2 + 2a_0a_1\cos(2\pi x/p), \quad \beta_0 = a_0^2 + a_1{}^2, \quad \beta_1 = 2a_0a_1, \quad \beta_2 = 0$$

$$NILS = -\pi\frac{\beta_1}{\beta_0 - \beta_2} = -\pi\frac{2a_0a_1}{a_0^2 + a_1{}^2}.$$ For equal liens and spaces, $a_0 = 1/2$, $a_1 = 1/\pi$.

$$NILS = -\frac{1}{\frac{1}{4} + \frac{1}{\pi^2}} = -\frac{4\pi^2}{4 + \pi^2} \approx 2.85$$

3.14. For the line/space image of equation (3.86):

a) derive an equation for the image contrast, defined as

$$Image\ Contrast = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

For simplicity, assume the maximum intensity occurs in the middle of the space, and the minimum intensity occurs in the middle of the line.

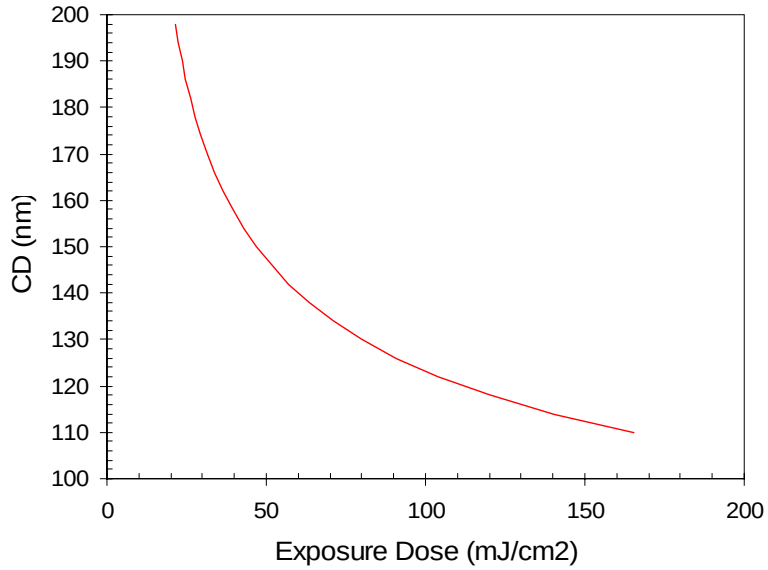
b) For the case where the second harmonic is the largest harmonic in the image (i.e., when $N = 2$), compare the resulting image contrast to the NILS of equation (3.89).

a) $I(x) = \sum_{j=0}^{N}\beta_j\cos(2\pi jx/p)$, $I_{\max} = I(x=0)$, $I_{\min} = I(x = p/2)$

$$I_{\max} = \sum_{j=0}^{N}\beta_j, \quad I_{\min} = \sum_{j=0}^{N}(-1)^j\beta_j, \quad Image\ Contrast = \frac{\beta_1 + \beta_3 + \beta_5 + \ldots}{\beta_0 + \beta_2 + \beta_4 + \ldots}$$

b) $Image\ Contrast = \dfrac{\beta_1}{\beta_0 + \beta_2}$, $NILS = -\pi\dfrac{\beta_1}{\beta_0 - \beta_2}$. Difference is that the second harmonic (β_2) improves the NILS but makes the image contrast worse.

3.15. Consider an in-focus, coherent, three-beam image of an equal line/space pattern under TE illumination (NA = 0.93, λ = 193 nm, linewidth = 130 nm, pitch = 260 nm). Use a threshold model (E_{th} = 20 mJ/cm^2) to plot the resulting line CD versus dose. What dose produces the nominal feature size?



Chapter 4

4.1. Imagine two plane waves moving in opposite directions along the *z*-axis. Using the time-dependent form of the waves rather than the phasor representation, the +*z* moving wave is given by

$$U_+(z,t) = A\cos(\omega t - kz)$$

and the –*z* moving wave is

$$U_-(z,t) = A\cos(\omega t + kz)$$

Show that the sum of these waves is separable in *z* and *t* (that is, can be expressed as product of a function of *z* and a function of *t*). Show that this results in an intensity 'standing wave', that is, the time average of the square of the electric field varies sinusoidally with position.

Use the trig identity $\cos(a+b) = \cos a \cos b - \sin a \sin b$:

$$U_+ + U_- = A\left[\cos(\omega t - kz) + \cos(\omega t + kz)\right] = 2A\cos(\omega t)\cos(kz)$$

$$(U_+ + U_-)^2 = 4A^2\cos^2(\omega t)\cos^2(kz) = 4A^2\cos^2(kz)\left(\frac{1}{2} + \frac{1}{2}\cos(2\omega t)\right)$$

Taking the time average, the $\cos(2\omega t)$ term will average to zero, giving

$$I = 4A^2\cos^2(\omega t)\cos^2(kz) = 2A^2\cos^2(kz)$$

4.2. Which of the following situations will result in the formation of standing waves? Explain your reasoning.

(a) the superposition of identical waves that travel in the same direction
(b) the superposition of identical waves that travel in opposite directions
(c) the superposition of waves that are nearly identical but of different amplitudes that travel in opposite directions
(d) the superposition of nearly identical waves of slightly different frequencies that travel in the same direction

a) No. The result will be a traveling wave of twice the amplitude.
b) Yes. See problem 4.1.
c) Yes. This is just like light reflecting off a non-ideal mirror.
d) No. It is a traveling wave with beat frequencies (see below).

Use the trig identity $\cos(a) + \cos(b) = 2\cos\left[(a+b)/2\right]\cos\left[(a-b)/2\right]$:

$$U_1 + U_2 = \cos(\omega_1 t - kz) + \cos(\omega_2 t - kz) = 2\cos(\omega_{avg} t - kz)\cos(\omega_{beat} t - kz)$$

where $\omega_{avg} = (\omega_1 + \omega_2)/2$ and $\omega_{beat} = (\omega_1 - \omega_2)/2$.

4.3. Derive equation (4.19).

Start with $\dfrac{E_T(z)}{E_I} = \dfrac{\tau_{12}\left(e^{i2\pi \boldsymbol{n}_2 z/\lambda} + \rho_{23}\tau_D^2 e^{-i2\pi \boldsymbol{n}_2 z/\lambda}\right)}{1+\rho_{12}\rho_{23}\tau_D^2}$ and use $I(z) = \dfrac{n_2\left|E(z)\right|^2}{n_1\left|E_I\right|^2}$.

Defining $T_{eff} = \dfrac{\dfrac{n_2}{n_1}\left|\tau_{12}\right|^2}{\left|1+\rho_{12}\rho_{23}\tau_D^2\right|^2}$, it is obvious that $I(z) = T_{eff}\left|e^{i2\pi \boldsymbol{n}_2 z/\lambda} + \rho_{23}\tau_D^2 e^{-i2\pi \boldsymbol{n}_2 z/\lambda}\right|^2$.

Let $\rho_{23} = \left|\rho_{23}\right| e^{i\phi_{23}}$ and use the definition $\tau_D = e^{i2\pi \boldsymbol{n}_2 D/\lambda}$, giving $I(z) = T_{eff}\left|e^{i2\pi \boldsymbol{n}_2 z/\lambda} + \left|\rho_{23}\right| e^{-i(2\pi \boldsymbol{n}_2\{z-2D\}/\lambda - \phi_{23})}\right|^2$. Since we will need to separate out real and imaginary parts, use the complex refractive index $\boldsymbol{n_2} = n_2 + i\kappa$ and $\alpha = 4\pi\kappa/\lambda$.

$$I(z) = T_{eff}\left|e^{-\alpha z/2} e^{i2\pi n_2 z/\lambda} + e^{-\alpha\{2D-z\}/2}\left|\rho_{23}\right| e^{-i(2\pi n_2\{z-2D\}/\lambda - \phi_{23})}\right|^2$$

$$I(z) = T_{eff} e^{-\alpha z}\left|e^{i2\pi n_2 z/\lambda} + e^{-\alpha(D-z)}\left|\rho_{23}\right| e^{-i(2\pi n_2\{z-2D\}/\lambda - \phi_{23})}\right|^2$$

Now use Euler's identity to change the complex exponentials into rectangular form and collect up real and imaginary parts.

$$\text{Real} = \cos\left(2\pi n_2 z/\lambda\right) + \left|\rho_{23}\right| e^{-\alpha(D-z)}\cos\left(2\pi n_2\{z-2D\}/\lambda - \phi_{23}\right)$$

$$\text{Im} = \sin\left(2\pi n_2 z/\lambda\right) - \left|\rho_{23}\right| e^{-\alpha(D-z)}\sin\left(2\pi n_2\{z-2D\}/\lambda - \phi_{23}\right)$$

Square and sum them to get the magnitude:

$$\text{Mag} = 1 + \left|\rho_{23}\right|^2 e^{-\alpha 2(D-z)} + 2\left|\rho_{23}\right| e^{-\alpha(D-z)} A$$

where $A = \cos\left(2\pi n_2\{z-2D\}/\lambda - \phi_{23}\right)\cos\left(2\pi n_2 z/\lambda\right) - \sin\left(2\pi n_2\{z-2D\}/\lambda - \phi_{23}\right)\sin\left(2\pi n_2 z/\lambda\right)$

Using the trig identity $\cos(a+b) = \cos a\cos b - \sin a\sin b$,

$A = \cos\left(4\pi n_2\{z-D\}/\lambda - \phi_{23}\right)$

Thus,

$$I(z) = T_{eff} e^{-\alpha z}\left[1 + \left|\rho_{23}\right|^2 e^{-\alpha 2(D-z)} + 2\left|\rho_{23}\right| e^{-\alpha(D-z)}\cos\left(4\pi n_2\{z-D\}/\lambda - \phi_{23}\right)\right]$$

$$I(z) = T_{eff}\left[e^{-\alpha z} + \left|\rho_{23}\right|^2 e^{-\alpha(2D-z)} + 2\left|\rho_{23}\right| e^{-\alpha D}\cos\left(4\pi n_2\{D-z\}/\lambda + \phi_{23}\right)\right]$$

QED.

4.4. The *optical admittance* of a material, η, at normal incidence is just its refractive index, $\boldsymbol{n}$. For light traveling through that material at an angle θ, the optical admittance of the material for TE (*s*-polarized) and TM (*p*-polarized) waves are

$$\eta_{TE} = \boldsymbol{n}\cos\theta \qquad \eta_{TM} = \frac{\boldsymbol{n}}{\cos\theta}$$

Show that by using optical admittance in the Fresnel formulas (4.29), one obtains a single equation for reflection for both polarizations that has the same form as the normally incident definition of equation (4.8). What are the results for transmittance and how do they compare to equation (4.7)?

Fresnel formulas:

TE: $\rho_{ij\perp}(\theta_j) = \dfrac{\boldsymbol{n}_i\cos(\theta_i) - \boldsymbol{n}_j\cos(\theta_j)}{\boldsymbol{n}_i\cos(\theta_i) + \boldsymbol{n}_j\cos(\theta_j)} = \dfrac{\eta_i - \eta_j}{\eta_i + \eta_j}$

TM: $\rho_{ij\|}(\theta_j) = \dfrac{\boldsymbol{n}_i\cos(\theta_j) - \boldsymbol{n}_j\cos(\theta_i)}{\boldsymbol{n}_i\cos(\theta_j) + \boldsymbol{n}_j\cos(\theta_i)} = \dfrac{\dfrac{\boldsymbol{n}_i}{\cos(\theta_i)} - \dfrac{\boldsymbol{n}_j}{\cos(\theta_j)}}{\dfrac{\boldsymbol{n}_i}{\cos(\theta_i)} + \dfrac{\boldsymbol{n}_j}{\cos(\theta_j)}} = \dfrac{\eta_i - \eta_j}{\eta_i + \eta_j}$

Thus, using the optical admittance, both the TE and TM reflection coefficients become identical to the form of the normally incident reflection coefficient.

TE: $\tau_{ij\perp}(\theta_j) = \dfrac{2\boldsymbol{n}_i\cos(\theta_i)}{\boldsymbol{n}_i\cos(\theta_i) + \boldsymbol{n}_j\cos(\theta_j)} = \dfrac{2\eta_i}{\eta_i + \eta_j}$

TM: $\tau_{ij\|}(\theta_j) = \dfrac{2\boldsymbol{n}_i\cos(\theta_i)}{\boldsymbol{n}_i\cos(\theta_j) + \boldsymbol{n}_j\cos(\theta_i)} = \dfrac{\cos(\theta_i)}{\cos(\theta_j)}\dfrac{2\dfrac{\boldsymbol{n}_i}{\cos(\theta_i)}}{\dfrac{\boldsymbol{n}_i}{\cos(\theta_i)} + \dfrac{\boldsymbol{n}_j}{\cos(\theta_j)}} = \dfrac{\cos(\theta_i)}{\cos(\theta_j)}\dfrac{2\eta_i}{\eta_i + \eta_j}$

The TE transmittance looks like the normally incident case using optical admittances. For the TM case, we get the same normally incident form multiplied by the ratio of projected areas of incident and transmitted beams.

4.5. *Brewster's angle* is defined as the angle at which *p*-polarized light will have zero reflectance at the interface between two materials. Derive an equation for Brewster's angle. For *p*-polarized light incident on a photoresist-coated substrate at Brewster's angle, what is the resulting swing amplitude?

TM: $\rho_{ij\|}(\theta_j) = \dfrac{\boldsymbol{n}_i\cos(\theta_j) - \boldsymbol{n}_j\cos(\theta_i)}{\boldsymbol{n}_i\cos(\theta_j) + \boldsymbol{n}_j\cos(\theta_i)} = 0$, so $\boldsymbol{n}_i\cos(\theta_j) = \boldsymbol{n}_j\cos(\theta_i)$. Using Snell's law to relate incident and transmitted angles, $\boldsymbol{n}_i\sin(\theta_i) = \boldsymbol{n}_j\sin(\theta_j)$. Cross-multiplying these equations gives $\boldsymbol{n}_i\boldsymbol{n}_j\sin(\theta_i)\cos(\theta_i) = \boldsymbol{n}_i\boldsymbol{n}_j\sin(\theta_j)\cos(\theta_j)$ and $\sin(\theta_i)\cos(\theta_i) = \sin(\theta_j)\cos(\theta_j)$. Using the double angle formula for the sine gives $\sin(2\theta_i) = \sin(2\theta_j)$. This equation is trivially true if incident and transmitted angles are the same, but that solution does not satisfy Snell's law. It is also true when $\theta_i + \theta_j = \pi/2$. Using this result to eliminate θ_j from Snell's law,

$\boldsymbol{n}_i \sin(\theta_i) = \boldsymbol{n}_j \sin(\pi/2 - \theta_i) = \boldsymbol{n}_j \cos(\theta_i)$, so that

$\tan(\theta_i) = \dfrac{\boldsymbol{n}_j}{\boldsymbol{n}_i}$.

The swing amplitude is $4|\rho_{12}\rho_{23}|e^{-\alpha D}$. But since at Brewster's angle, $\rho_{12} = 0$ for *p*-polarized light. Thus, the swing amplitude will be zero for this case.

4.6. For *p*-polarized light traveling at 45º with respect to the normal inside of a photoresist, no standing waves will occur since the incident and reflected waves will have no overlap in their electric fields [see equation (4.37)]. Will the resist swing curve similarly have zero amplitude at this angle? What if the light is traveling at 45º with respect to the normal in air? Explain.

The swing curve is caused by interference between rays traveling in the same direction (reflected from the photoresist surface). Thus, regardless of polarization and angle, the electric fields of all the interfering rays will be 100% overlapped and there will be perfect interference. Other than the difference in transmission and reflection coefficients with angle and polarization, there will be no difference between TE and TM swing curves. Of course, the TM swing curve goes to zero at Brewster's angle (see problem 4.5), but this has nothing to do with the amount of overlap of the electric fields. For a typical resist, Brewster's angle is about 60° in air, and about 30° in the resist.

4.7. Derive equation (4.46) from equation (4.45).

$\rho_{total} = \dfrac{\rho_{12} + \rho_{23}\tau_D^2}{1 + \rho_{12}\rho_{23}\tau_D^2}$, $R = |\rho_{total}|^2$. Let $\rho_{23} = |\rho_{23}|e^{i\phi_{23}}$, $\rho_{12} = |\rho_{12}|e^{i\phi_{12}}$, and $\tau_D = e^{i2\pi \boldsymbol{n}_2 D/\lambda}$.

Numerator:

$$\rho_{12} + \rho_{23}\tau_D^2 = |\rho_{12}|e^{i\phi_{12}} + |\rho_{23}|e^{i\phi_{23}}e^{i2\pi \boldsymbol{n}_2 D/\lambda} = e^{i\phi_{12}}\left(|\rho_{12}| + |\rho_{23}|e^{i(2\pi \boldsymbol{n}_2 D/\lambda + \phi_{23} - \phi_{12})}\right)$$

$$\left|\rho_{12} + \rho_{23}\tau_D^2\right|^2 = |\rho_{12}|^2 + |\rho_{23}|^2 + 2|\rho_{12}\rho_{23}|\cos(2\pi \boldsymbol{n}_2 D/\lambda + \phi_{23} - \phi_{12})$$

Denominator:

$$1 + \rho_{12}\rho_{23}\tau_D^2 = 1 + |\rho_{12}|e^{i\phi_{12}}|\rho_{23}|e^{i\phi_{23}}e^{i2\pi \boldsymbol{n}_2 D/\lambda} = 1 + |\rho_{12}\rho_{23}|e^{i(2\pi \boldsymbol{n}_2 D/\lambda + \phi_{23} + \phi_{12})}$$

$$\left|1 + \rho_{12}\rho_{23}\tau_D^2\right|^2 = 1 + |\rho_{12}\rho_{23}|^2 + 2|\rho_{12}\rho_{23}|\cos(2\pi \boldsymbol{n}_2 D/\lambda + \phi_{23} + \phi_{12})$$

Thus,

$$R = \frac{|\rho_{12}|^2 + |\rho_{23}|^2 + 2|\rho_{12}\rho_{23}|\cos(2\pi \boldsymbol{n}_2 D/\lambda + \phi_{23} - \phi_{12})}{1 + |\rho_{12}\rho_{23}|^2 + 2|\rho_{12}\rho_{23}|\cos(2\pi \boldsymbol{n}_2 D/\lambda + \phi_{23} + \phi_{12})}$$

Alternate (easier) derivation: Use the property that $|a+b|^2 = |a|^2 + |b|^2 + 2|ab|\cos(\theta_a - \theta_b)$ for any two complex numbers *a* and *b*.

4.8. Derive an expression for the CD swing curve using equations (4.51) – (4.53).

Equation (4.51): $E = E_{crit} \dfrac{1 + \left|\rho_{12}\rho_{23}\right|^2 e^{-\alpha 2D} + 2\left|\rho_{12}\rho_{23}\right| e^{-\alpha D} \cos(4\pi n_2 D/\lambda + \phi_{12} + \phi_{23})}{T_{12}\left(1 + \left|\rho_{23}\right|^2\right) e^{-\alpha D}}$

Equation (4.52): $\dfrac{CD - CD_1}{CD_1} = \left.\dfrac{d \ln CD}{d \ln E}\right|_{E=E_1} \left(1 - \dfrac{E_1}{E}\right)$

Equation (4.53): $E_1 = E_{crit} \dfrac{1 + \left|\rho_{12}\rho_{23}\right|^2 e^{-\alpha 2D_{\min}} - 2\left|\rho_{12}\rho_{23}\right| e^{-\alpha D_{\min}}}{T_{12}\left(1 + \left|\rho_{23}\right|^2\right) e^{-\alpha D_{\min}}}$

Let $s = \left.\dfrac{d \ln CD}{d \ln E}\right|_{E=E_1}$, and assume that it is constant.

$$\frac{CD - CD_1}{CD_1} = s\left(1 - \frac{e^{-\alpha(D - D_{\min})}\left(1 + \left|\rho_{12}\rho_{23}\right|^2 e^{-\alpha 2D_{\min}} - 2\left|\rho_{12}\rho_{23}\right| e^{-\alpha D_{\min}}\right)}{1 + \left|\rho_{12}\rho_{23}\right|^2 e^{-\alpha 2D} + 2\left|\rho_{12}\rho_{23}\right| e^{-\alpha D} \cos(4\pi n_2 D/\lambda + \phi_{12} + \phi_{23})}\right)$$

4.9. Derive equation (4.84).

Derive $D = \dfrac{\lambda}{4\pi\kappa_2} \ln\left|\dfrac{\rho_{23}}{\rho_{21}}\right| = \dfrac{\lambda}{4\pi n_2}\left(\theta_{21} - \theta_{23}\right)$.

These two equation come from setting $\rho_{21} = \rho_{23}\tau_D^2$ and then equating both the magnitude and phase of each side of the equation.

$$\left|\rho_{21}\right| e^{i\theta_{21}} = \left|\rho_{23}\right| e^{i\theta_{23}} e^{i4\pi(n_2 + i\kappa_2)D/\lambda} = \left|\rho_{23}\right| e^{-4\pi\kappa_2 D/\lambda} e^{i(4\pi n_2 D/\lambda + \theta_{23})}$$

Magnitude: $\left|\rho_{21}\right| = \left|\rho_{23}\right| e^{-4\pi\kappa_2 D/\lambda}$, $-\dfrac{4\pi\kappa_2 D}{\lambda} = \ln\left|\dfrac{\rho_{21}}{\rho_{22}}\right|$, $D = \dfrac{\lambda}{4\pi\kappa_2} \ln\left|\dfrac{\rho_{23}}{\rho_{21}}\right|$

Phase: $\theta_{21} = 4\pi n_2 D/\lambda + \theta_{23}$, $D = \dfrac{\lambda}{4\pi n_2}\left(\theta_{21} - \theta_{23}\right)$

4.10. Calculate the swing ratio for a nominally 400-nm-thick 193-nm resist (assume typical properties) on a substrate of copper. What is the impact on this swing ratio if the maximum is at a greater thickness than the minimum, compared to the minimum at a greater thickness than the maximum?

$$SR = \frac{e^{\alpha D}\left(e^{\alpha\Delta D/2} - e^{-\alpha\Delta D/2}\right) + \left|\rho_{12}\rho_{23}\right|^2 e^{-\alpha D}\left(e^{-\alpha\Delta D/2} - e^{\alpha\Delta D/2}\right) + 4\left|\rho_{12}\rho_{23}\right|}{\frac{1}{2}\left[e^{\alpha D}\left(e^{\alpha\Delta D/2} + e^{-\alpha\Delta D/2}\right) + \left|\rho_{12}\rho_{23}\right|^2 e^{-\alpha D}\left(e^{-\alpha\Delta D/2} + e^{\alpha\Delta D/2}\right)\right]}$$

Approximate: $SR \approx \alpha\Delta D + 4\left|\rho_{12}\rho_{23}\right| e^{-\alpha D}$

From Table 4.3, $n_2 = 1.71 + i0.018$, $n_3 = 0.970 + i1.40$ (at 193 nm). This gives

$$\alpha = \frac{4\pi\kappa_2}{\lambda} = 1.17\ \mu\text{m}^{-1}$$

Period = $\Delta D = \dfrac{\lambda}{2n_2} = 56.4$ nm

$$\left|\rho_{ij}\right| = \sqrt{\frac{\left(n_i - n_j\right)^2 + \left(\kappa_i - \kappa_j\right)^2}{\left(n_i + n_j\right)^2 + \left(\kappa_i + \kappa_j\right)^2}}, \quad \left|\rho_{12}\right| = \sqrt{\frac{\left(1 - 1.71\right)^2 + \left(0.018\right)^2}{\left(1 + 1.71\right)^2 + \left(0.018\right)^2}} = 0.262$$

$$\left|\rho_{23}\right| = \sqrt{\frac{\left(1.71 - 0.97\right)^2 + \left(0.018 - 1.4\right)^2}{\left(1.71 + 0.97\right)^2 + \left(0.018 + 1.4\right)^2}} = 0.517$$

$$\left|\rho_{12}\rho_{23}\right|^2 e^{-\alpha D} = 0.0129$$

$$SR = \frac{1.42\left(1.0335 - 0.9675\right) + 0.0129\left(-1.0335 + 0.9675\right) + 4(0.262)(0.517)}{\frac{1}{2}\left[1.42\left(1.0335 + 0.9675\right) + 0.0129\left(1.0335 + 0.9675\right)\right]} = 0.443$$

Approximate: $SR \approx \alpha\Delta D + 4\left|\rho_{12}\rho_{23}\right| e^{-\alpha D} = 1.17(0.0564) + 4(0.262)(0.517)(0.704) = 0.447$

Thus, even for a high-absorbance 193 nm resist, the approximate expression is good enough (1% error)!

Changing the sign of ΔD, $SR = 0.315$ (using the approximate expression).

4.11. Using the normal-incidence BARC solution given in Figure 4.16 for a 40-nm BARC thickness, What BARC thickness control is required to keep the BARC intensity reflectivity below 0.5%?

From the figure, at 40 nm, $n_2 = 1.53 + i0.5$, $n_1 = 1.7 + i0.015$.

$$R \approx \Delta^2\left(\frac{\pi n_1}{\lambda}\right)^2\left|1 - \left(\frac{\boldsymbol{n}_2}{\boldsymbol{n}_1}\right)^2\right|^2$$

Ignoring the imaginary part of n_1,

$$R \approx \Delta^2\left(\frac{\pi n_1}{\lambda}\right)^2\left|1 - \frac{n_2^2 - \kappa_2^2 + i2n_2\kappa_2}{n_1^2}\right|^2 = \Delta^2\left(\frac{\pi}{\lambda}\right)^2 \frac{\left(n_1^2 - n_2^2 + \kappa_2^2\right)^2 + 4n_2^2\kappa_2^2}{n_1^2}$$

$$R = \Delta^2 \left(\frac{\pi}{\lambda}\right)^2 \left(n_1^{\,2} - 2\left(n_2^{\,2} - \kappa_2^{\,2}\right) + \frac{\left(n_2^{\,2} + \kappa_2^{\,2}\right)^2}{n_1^{\,2}} \right)$$

$$0.005 = \Delta^2 \left(\frac{\pi}{193}\right)^2 (1.031), \quad \Delta = \pm 4.3 \text{ nm.}$$

4.12. At 248 nm, a typical resist on silicon will result in a swing amplitude of about 0.69. Using a BARC with a reflectivity of 1%, the swing amplitude is reduced to about 0.095. Ignoring absorption, what level of resist thickness control is required for each of these two cases (resist on silicon and resist on a 1% reflective BARC) to keep the effective dose error below 1%.

$\frac{\Delta E}{E} \approx \left(\frac{\alpha\Delta + 4\left|\rho_{12}\rho_{23}\right| e^{-\alpha D}\left(2\pi n_2/\lambda\right)^2}{1 - 2\left|\rho_{12}\rho_{23}\right| e^{-\alpha D}} \right)$. Ignoring absorption and for small thickness errors, we see that $\frac{\Delta E}{E} \approx \left(\frac{(swing\ amplitude)\left(2\pi n_2 \Delta/\lambda\right)^2}{1 - (swing\ amplitude)/2} \right)$.

At 248 nm, $n_2 = 1.76$ and $\left(2\pi n_2/\lambda\right)^2 = 0.001988\ \text{nm}^{-1}$.

On silicon, $0.01 = \frac{0.69(0.001988)}{1 - 0.69/2}\Delta^2$ and $\Delta = 2.5$ nm.

On BARC, $0.01 = \frac{0.095(0.001988)}{1 - .095/2}\Delta^2$ and $\Delta = 7.1$ nm.

4.13. Derive equation (4.103), the refractive index of an ideal TARC.

For the ideal TARC, $\rho_{12} = \rho_{23}$ and all refractive indices are real.

$$\frac{n_1 - n_2}{n_1 + n_2} = \frac{n_2 - n_3}{n_2 + n_3}, \quad \left(n_1 - n_2\right)\left(n_2 + n_3\right) = \left(n_1 + n_2\right)\left(n_2 - n_3\right)$$

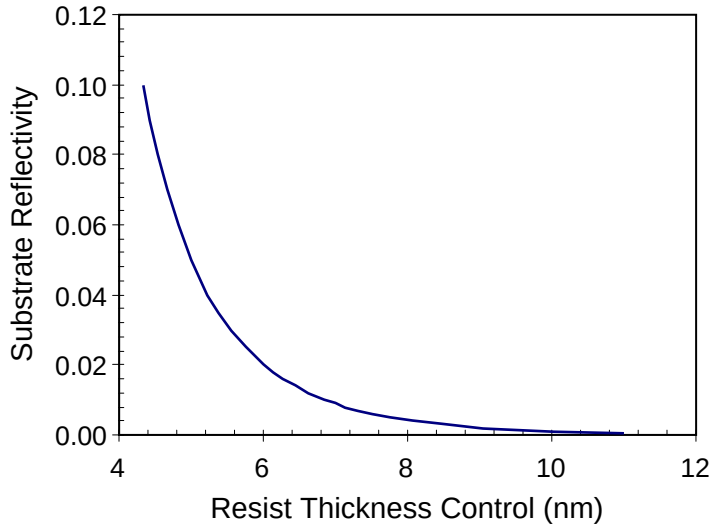
$$n_1 n_2 - n_2^{\,2} + n_1 n_3 - n_2 n_3 = n_1 n_2 + n_2^{\,2} - n_1 n_3 - n_2 n_3$$

$$-n_2^{\,2} + n_1 n_3 = n_2^{\,2} - n_1 n_3$$

$$2 n_1 n_3 = 2 n_2^{\,2}$$

$$n_2 = \sqrt{n_1 n_3}$$

4.14. As an example of the use of equation (4.100), consider a 193-nm resist with $n_2 = 1.7$, $\alpha = 1.2\ \mu\text{m}^{-1}$, $D = 200$nm, and $|\rho_{12}| = 0.26$. Suppose that 2% dose error due to resist thickness variations was considered acceptable. Generate a plot of maximum allowed substrate reflectivity ($|\rho_{23}|^2$) as a function of the resist thickness variation Δ.



4.15. What will be the impact of using immersion lithography (versus dry lithography at the same numerical aperture) in terms of the contrast of the image-in-resist for unpolarized illumination? Explain.

The angles of light inside the resist will be the same for the same numerical aperture (and the same feature being imaged), regardless of whether immersion lithography is used. Thus, the only impact will be on the transmission of light at large angles at the air/resist versus the fluid/resist interface. For dry lithography and a small pitch pattern, TM light transmits the first orders better than TE light, resulting in a slightly lower contrast in resist since the TM light reduces contrast. For immersion lithography, the transmission of TE and TM light is more equal, thus allowing a relatively greater amount of TE light into the resist, giving a slightly higher contrast.

4.16. Derive equation (4.117) from equation (4.116).

Starting with
$E_x = e^{ikz}\left(a_0 + 2a_1\cos\theta\cos(2\pi x/p)e^{-ikz(1-\cos\theta)}\right)$ and $E_z = ie^{ikz\cos\theta}\left(2a_1\sin\theta\sin(2\pi x/p)\right)$

Derive $I(x,z) = a_0{}^2 + 2a_1{}^2 + 4a_0a_1\cos\theta\cos(kz(1-\cos\theta))\cos(2\pi x/p) + 2a_1{}^2\cos(2\theta)\cos(4\pi x/p)$

Note that absorption is ignored in this result, so we will assume the resist propagation constant is real.

$$I(x,z) = |E_x|^2 + |E_z|^2$$
$$|E_x|^2 = \left[a_0 + 2a_1\cos\theta\cos(2\pi x/p)\cos\left(kz(1-\cos\theta)\right)\right]^2 + \left[2a_1\cos\theta\cos(2\pi x/p)\sin\left(kz(1-\cos\theta)\right)\right]^2$$
$$= a_0{}^2 + 4a_0a_1\cos\theta\cos(2\pi x/p)\cos\left(kz(1-\cos\theta)\right) + 4a_1{}^2\cos^2\theta\cos^2(2\pi x/p)$$
$$|E_z|^2 = 4a_1{}^2\sin^2\theta\sin^2(2\pi x/p)$$

Noting that $\sin^2(a)\sin^2(b) + \cos^2(a)\cos^2(b) = \frac{1}{2} + \frac{1}{2}\cos(2a)\cos(2b)$,

$$\left|E_x\right|^2 + \left|E_z\right|^2 = a_0{}^2 + 2a_1{}^2 + 4a_0 a_1 \cos\theta \cos(2\pi x/p)\cos\big(kz(1-\cos\theta)\big) + 2a_1{}^2 \cos 2\theta \cos(4\pi x/p)$$

4.17. For light incident on a plane interface between two materials, use equation (4.141) to show that $R + T = 1$ for each polarization.

$$R = \left|\rho_{12}\right|^2, \quad T = \left|\frac{\boldsymbol{n}_2 \cos(\theta_2)}{\boldsymbol{n}_1 \cos(\theta_1)}\right| \left|\tau_{12}\right|^2$$

TE: $$\rho_{ij\perp}(\theta_j) = \frac{\boldsymbol{n}_i \cos(\theta_i) - \boldsymbol{n}_j \cos(\theta_j)}{\boldsymbol{n}_i \cos(\theta_i) + \boldsymbol{n}_j \cos(\theta_j)}, \quad \tau_{ij\perp}(\theta_j) = \frac{2\boldsymbol{n}_i \cos(\theta_i)}{\boldsymbol{n}_i \cos(\theta_i) + \boldsymbol{n}_j \cos(\theta_j)}$$

$$R + T = \left|\frac{\boldsymbol{n}_1 \cos(\theta_1) - \boldsymbol{n}_2 \cos(\theta_2)}{\boldsymbol{n}_1 \cos(\theta_1) + \boldsymbol{n}_2 \cos(\theta_2)}\right|^2 + \left|\frac{\boldsymbol{n}_2 \cos(\theta_2)}{\boldsymbol{n}_1 \cos(\theta_1)}\right| \left|\frac{2\boldsymbol{n}_1 \cos(\theta_1)}{\boldsymbol{n}_1 \cos(\theta_1) + \boldsymbol{n}_2 \cos(\theta_2)}\right|^2$$

$$R + T = \left(\left|\boldsymbol{n}_1 \cos(\theta_1) - \boldsymbol{n}_2 \cos(\theta_2)\right|^2 + \left|\frac{\boldsymbol{n}_2 \cos(\theta_2)}{\boldsymbol{n}_1 \cos(\theta_1)}\right| \left|2\boldsymbol{n}_1 \cos(\theta_1)\right|^2\right) \left|\frac{1}{\boldsymbol{n}_1 \cos(\theta_1) + \boldsymbol{n}_2 \cos(\theta_2)}\right|^2$$

Note that $\left|A\right|^2 = \left|A^2\right|$. Thus,

$$\left|\boldsymbol{n}_1 \cos(\theta_1) - \boldsymbol{n}_2 \cos(\theta_2)\right|^2 + \left|\frac{\boldsymbol{n}_2 \cos(\theta_2)}{\boldsymbol{n}_1 \cos(\theta_1)}\right| \left|2\boldsymbol{n}_1 \cos(\theta_1)\right|^2 =$$

$$\left|\big(\boldsymbol{n}_1 \cos(\theta_1)\big)^2 + \big(\boldsymbol{n}_2 \cos(\theta_2)\big)^2 - 2\boldsymbol{n}_1 \boldsymbol{n}_2 \cos(\theta_1)\cos(\theta_2)\right| + 4\left|\frac{\boldsymbol{n}_2 \cos(\theta_2)}{\boldsymbol{n}_1 \cos(\theta_1)}\right| \left|\big(\boldsymbol{n}_1 \cos(\theta_1)\big)^2\right| =$$

$$\left|\big(\boldsymbol{n}_1 \cos(\theta_1)\big)^2\right| + \left|\big(\boldsymbol{n}_2 \cos(\theta_2)\big)^2\right| - \left|2\boldsymbol{n}_1 \boldsymbol{n}_2 \cos(\theta_1)\cos(\theta_2)\right| + 4\left|\big(\boldsymbol{n}_2 \cos(\theta_2)\boldsymbol{n}_1 \cos(\theta_1)\big)\right| =$$

$$\left|\big(\boldsymbol{n}_1 \cos(\theta_1)\big)^2\right| + \left|\big(\boldsymbol{n}_2 \cos(\theta_2)\big)^2\right| + \left|2\boldsymbol{n}_1 \boldsymbol{n}_2 \cos(\theta_1)\cos(\theta_2)\right| = \left|\boldsymbol{n}_1 \cos(\theta_1) + \boldsymbol{n}_2 \cos(\theta_2)\right|^2$$

Combining with the denominator gives $R + T = 1$.

TM: $$\rho_{ij\parallel}(\theta_j) = \frac{\boldsymbol{n}_i \cos(\theta_j) - \boldsymbol{n}_j \cos(\theta_i)}{\boldsymbol{n}_i \cos(\theta_j) + \boldsymbol{n}_j \cos(\theta_i)}, \quad \tau_{ij\parallel}(\theta_j) = \frac{2\boldsymbol{n}_i \cos(\theta_i)}{\boldsymbol{n}_i \cos(\theta_j) + \boldsymbol{n}_j \cos(\theta_i)}$$

$$R + T = \left|\frac{\boldsymbol{n}_1 \cos(\theta_2) - \boldsymbol{n}_2 \cos(\theta_1)}{\boldsymbol{n}_1 \cos(\theta_2) + \boldsymbol{n}_2 \cos(\theta_1)}\right|^2 + \left|\frac{\boldsymbol{n}_2 \cos(\theta_2)}{\boldsymbol{n}_1 \cos(\theta_1)}\right| \left|\frac{2\boldsymbol{n}_1 \cos(\theta_1)}{\boldsymbol{n}_1 \cos(\theta_2) + \boldsymbol{n}_2 \cos(\theta_1)}\right|^2$$

The derivation from here is essentially identical to that for TE light.

Chapter 5

5.1. Assuming a plane wave traveling through a uniform material, derive equation (5.5).

Let $U = Ae^{ikz}$ where $k = 2\pi \boldsymbol{n}/\lambda$ and $\boldsymbol{n} = n + i\kappa$. Thus, $U = Ae^{i2\pi(n+i\kappa)z/\lambda} = Ae^{i2\pi nz/\lambda}e^{-2\pi\kappa z/\lambda}$ and $I = |U|^2 = |A|^2 e^{-4\pi\kappa z/\lambda}$. This has the exact form of the integrated Lambert law of absorption where $I_0 = |A|^2$ and $\alpha = 4\pi\kappa/\lambda$.

5.2. For a resist film of a given thickness, there is one value of the absorption coefficient that maximizes the amount of light absorbed by the resist at the bottom. (This can be seen by looking at the extremes. When α is zero, all of the incident light makes it to the bottom, but none is absorbed. When α is infinite, no light makes it to the bottom, so none is absorbed there. Thus, in between these two extremes there must be a maximum.) For a resist of thickness *D*, and assuming a homogenous material and a non-reflecting substrate, what value of α maximizes the amount of light absorbed by the resist at the bottom?

The amount of light absorbed at the bottom of the resist is = $-\dfrac{dI}{dz} = \alpha I = \alpha I_0 e^{-\alpha D}$. Maximize this quantity, let $y = \alpha e^{-\alpha D}$ and set $\dfrac{dy}{d\alpha} = 0$. This gives $0 = -\alpha D e^{-\alpha D} + e^{-\alpha D}$, $0 = -\alpha D + 1$ and $\alpha = 1/D$ for maximum absorption.

5.3. An i-line resist has the following properties:

A = 0.85 μm^{-1}
B = 0.05 μm^{-1}
C = 0.018 cm^2/mJ
Refractive index = 1.72

The resist is coated to a thickness of 1.1 μm on a glass substrate optically matched to the photoresist.

(a) At the beginning of exposure, what percentage of the incident light makes it to the bottom of the resist?
(b) The photoresist is exposed so that 55% of the sensitizer is converted at the top of the resist. Give your best estimate of what percentage of the incident light makes it to the bottom of the resist at the end of the exposure. Explain why this estimate is not exact.

a) At the start of exposure, $\alpha = A + B$ = 0.90 μm^{-1}. $\dfrac{I(D)}{I(0)} = e^{-\alpha D} = e^{-(0.90)(1.1)} = 0.3716$. But,

$T_{12} = 1 - \left(\dfrac{n_2 - n_1}{n_2 + n_1}\right)^2 = 1 - \left(\dfrac{1.72 - 1}{1.72 + 1}\right)^2 = 0.930$, so that $\dfrac{I(D)}{I_{inc}} = (0.930)(0.3716) = 0.346$

b) Since B « A, assume B = 0 and use $m(z) = \dfrac{m(0)}{m(0) + [1 - m(0)]e^{-Az}}$. For $m(0) = 0.45$,

$m(z) = \dfrac{0.45}{0.45 + [0.55]e^{-(0.85)(1.1)}} = 0.676$. $\dfrac{I(z)}{I(0)} = \dfrac{m(z)}{m(0)}e^{-Az} = \dfrac{0.676}{0.45}e^{-(0.85)(1.1)} = 0.590$.

This estimate is not exact since B is small but not zero.

5.4. For most conventional (g- and i-line) resists, a_M » a_P. For this case, show that a measurement of bleachable absorption coefficient as a function of wavelength, $A(\lambda)$, can be used to determine how the exposure rate constant C varies with wavelength.

$A = (a_M - a_P)\, M_0 \approx a_M\, M_0$. Also, from equation (5.28), $C = \dfrac{\Phi\, a_M\, \lambda}{N_A\, h\, c}$ so that $C \approx A\dfrac{\Phi\, \lambda}{M_0 N_A\, h\, c}$. If we assume that the quantum efficiency doesn't vary much with wavelength, measuring $A(\lambda)$ and C at one wavelength can be combined with this equation to give $C(\lambda)$.

5.5. Derive equations (5.35).

Derive $m(z) = \dfrac{m(0)}{m(0) + [1 - m(0)]e^{-Az}}$ and $\dfrac{I(z)}{I(0)} = \dfrac{e^{-Az}}{m(0) + [1 - m(0)]e^{-Az}} = \dfrac{m(z)}{m(0)}e^{-Az}$.

Start with $z = \displaystyle\int_{m(z=0)}^{m(z)} \frac{dy}{y[A(1-y) - B\ln(y)]}$ and assume A » B. Thus, we have

$z = \displaystyle\int_{m(z=0)}^{m(z)} \frac{dy}{A(1-y)y} = \frac{1}{A}\ln\left(\frac{y}{y-1}\right)\Bigg|_{m(0)}^{m(z)}$ and $e^{Az} = \left(\dfrac{m(z)}{m(z)-1}\right)\left(\dfrac{m(0)-1}{m(0)}\right)$. Solving for $m(z)$ gives

$m(z) = \dfrac{m(0)}{m(0) + [1 - m(0)]e^{-Az}}$.

$I(z) = I(0)\dfrac{A[1 - m(z)] - B\ln[m(z)]}{A[1 - m(0)] - B\ln[m(0)]}$ becomes $I(z) = I(0)\dfrac{1 - m(z)}{1 - m(0)}$. Using the above expression for $m(z)$, $1 - m(z) = 1 - \dfrac{m(0)}{m(0) + [1 - m(0)]e^{-Az}} = \dfrac{[1 - m(0)]e^{-Az}}{m(0) + [1 - m(0)]e^{-Az}}$ and

$\dfrac{I(z)}{I(0)} = \dfrac{e^{-Az}}{m(0) + [1 - m(0)]e^{-Az}}$.

5.6. Reciprocity during exposure is seen for the case of a nonbleaching resist by transforming equation (5.29) to equation (5.31) – all instances of incident intensity and exposure time can be replaced by incident dose.

(a) Show that the first equation (5.35) obeys reciprocity.
(b) Show that the general first-order exposure equation (5.27) results in reciprocity regardless of the effects of bleaching.

a) Equation (5.35): $m(z) = \dfrac{m(0)}{m(0) + \left[1 - m(0)\right] e^{-Az}}$ and $\dfrac{I(z)}{I(0)} = \dfrac{e^{-Az}}{m(0) + \left[1 - m(0)\right] e^{-Az}}$. In both of these expressions, exposure enters only through the $m(0)$ term. And since the $m(0)$ term obeys reciprocity (no effects of bleaching are seen at the top of the resist), both of these equations must obey reciprocity as well.

b) For the general bleaching case, $m = e^{-C(Dose)}$ where $Dose = \int_0^t I\,dt$. Using the definition of incident dose, $Dose = \int_0^t I\,dt = \int_0^t I_r I_{inc}\,dt = \int_0^{Dose_{inc}} I_r\,d(Dose_{inc})$ which is a function of incident dose only.

5.7 Suppose that a resist was found to produce the same thickness after PAB for a 95 ºC, 300-s bake, a 100 ºC, 60-s bake, or a 105 ºC, 17-s bake. If the DNQ of that resist decomposes with an activation energy of 31 Kcal/mole and ln(A_r) = 37 (1/min), which bake is preferable from a thermal decomposition perspective?

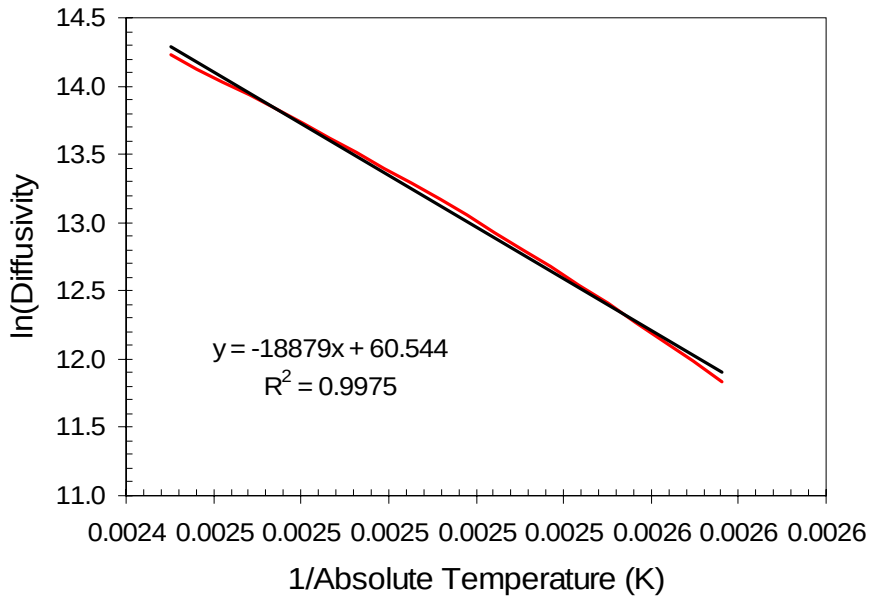
$m' = e^{-K_T t_b}$ and $K_T = A_r e^{-E_a / RT}$.

At T = 95 °C, t_b = 300 s, K_T = exp[37-31,000/1.98717/(273.15+95)] = 0.0046345 1/min, and m' = 0.977.

At T = 100 °C, t_b = 60 s, K_T = exp[37-31,000/1.98717/(273.15+100)] = 0.0081769 1/min, and m' = 0.992.

Thus, the higher temperature, shorter time bake produces less PAC decomposition.

5.8 The modified Fujita–Doolittle equation (5.61) predicts a temperature dependence for solvent diffusivity which is not the Arrhenius equation. Using the parameters from Table 5.1 and assuming 10% solvent volume fraction, create an Arrhenius plot (log[diffusivity] vs. 1/absolute temperature) for solvent diffusivity from 115 to 130 ºC. How close is this to true Arrhenius behavior? Find a best estimate for A_r and E_a based on this plot.



Obviously the plot is not exactly linear, but it is reasonably close. The Arrhenius equation is

$D = A_r e^{-E_a/RT}$ or $\ln(D) = \ln(A_r) - E_a/RT$, so from the linear fit we have $\ln(A_r) = 60.54$ (units of $\ln(\text{nm}^2/\text{s})$) and $E_a = 37.52$ kcal/mol.

5.9. Solve the 1D diffusion equation for a 'fixed source' boundary condition (diffusion of a constant concentration of material at the surface of a very thick resist). Assume the diffusivity is constant and

$$C(z,0) = 0$$
$$C(0,t) = C_0$$
$$C(\infty,t) = 0$$

Solution: $C(z,t) = C_0 \operatorname{erfc}\left(\dfrac{z}{\sqrt{2}\sigma_D}\right)$

5.10. Solve the 1D diffusion equation for the 'initial source' boundary condition (diffusion of a given amount material at the surface of a very thick resist). Assume the diffusivity is constant and

$$C(z,0) = 0$$
$$\mathrm{d}C(0,t)/\mathrm{d}t = 0$$
$$C(\infty,t) = 0$$
$$\int_0^\infty C(z,t)dz = Q_0$$

Solution: $C(z,t) = \sqrt{\dfrac{2}{\pi}}\dfrac{Q_0}{\sigma_D}e^{-z^2/2\sigma_D^2}$. Note that the surface concentration is $\sqrt{\dfrac{2}{\pi}}\dfrac{Q_0}{\sigma_D}$ and changes with bake time (diffusion length).

5.11. For a typical 300-mm silicon wafer baked at a hotplate temperature of 100 ºC (ambient temperature of 20 ºC), and assuming a proximity gap of 0.2 mm, calculate:

(a) The equilibrium wafer temperature (T^*) and the rise time of the bake (τ).
(b) The equilibrium wafer temperature and the rise time of the bake if the proximity gap were increased to 0.3 mm.
(c) At what proximity gap distance would the coefficient of conduction (heat transfer from the hotplate across the gap) equal that of convection (thermal transfer from the top of the wafer)?

a) Assume that the ambient temperature is 20 °C, use the value of k_{air} at 100 °C, and pick a value of h in the middle of the range, 7.5:

$$T^* = \frac{\frac{k_{air}}{\delta}T_{plate} + hT_{ambient}}{\frac{k_{air}}{\delta} + h} = \frac{\frac{0.031}{0.0002}(100) + 7.5(20)}{\frac{0.031}{0.0002} + 7.5} = 96.31\ °\text{C}$$

Alternately, $T^* \approx T_{plate} - \frac{\delta h}{k_{air}}\left(T_{plate} - T_{ambient}\right)$ = 96.13 °C (a pretty accurate approximation, especially given the uncertainty in h).

Assume that C_p = 21 J/K-mole, and convert to specific heat capacity by dividing by the atomic weight.

$$\tau \approx \frac{\delta \rho C_p L}{k_{air}} = \frac{(0.0002m)(2{,}330{,}000\,g/m^3)(21J/K-mol)(0.000775m)}{(0.031W/m-K)(28.09g/mol)} = 8.7\text{ s}$$

b) T^* = 94.2 °C, τ = 13.1 s.

c) $\frac{k_{air}}{\delta} = h$, or $\delta = \frac{k_{air}}{h} = \frac{0.031}{7.5} = 0.004m$ = 4 mm

5.12. For a typical 300-mm silicon wafer baked at a hotplate temperature of 100 ºC and a proximity gap of 0.2 mm, the equilibrium wafer temperature was found to vary by ±0.2 °C across the wafer. How much wafer warpage would be needed to explain all of this temperature variation?

$$T^* \approx T_{plate} - \frac{\delta h}{k_{air}}\left(T_{plate} - T_{ambient}\right) = 100 - \delta\frac{7.5}{0.031}(80) = 100 - \delta(19.35°C/mm)\ .$$

Thus, $\Delta\delta$ = 0.2/19.35 = 0.01 mm

5.13. Derive equation (5.79).

$$T^* = \frac{\frac{k_{air}}{\delta}T_{plate} + hT_{ambient}}{\frac{k_{air}}{\delta} + h},\quad when\ \frac{k_{air}}{\delta} >> h,\quad T^* \approx T_{plate} - \frac{\delta h}{k_{air}}\left(T_{plate} - T_{ambient}\right)$$

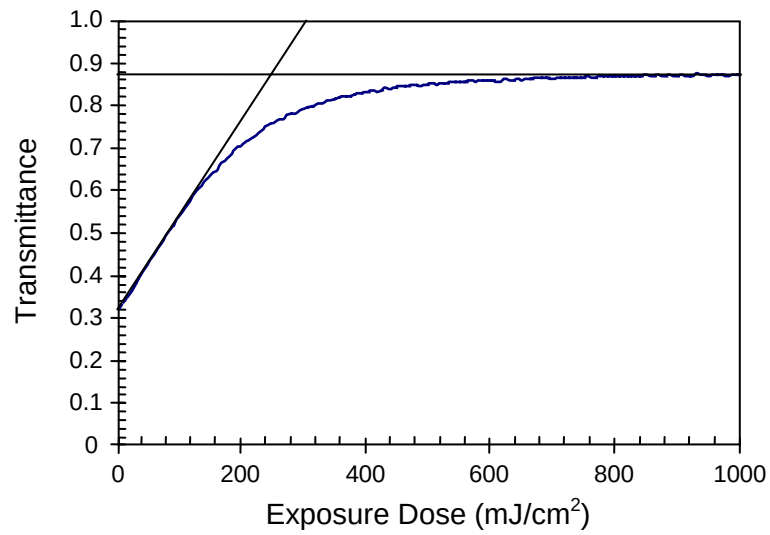
Factor out k_{air}/δ from both numerator and denominator.

$T^* = \dfrac{T_{plate} + \dfrac{h\delta}{k_{air}} T_{ambient}}{1 + \dfrac{h\delta}{k_{air}}}$. Use the series expansion $\dfrac{1}{1+r} \approx 1 - r$ for r « 1. Keeping only first-order terms,

$$T^* \approx \left(T_{plate} + \frac{h\delta}{k_{air}} T_{ambient} \right)\left(1 - \frac{h\delta}{k_{air}} \right) \approx T_{plate} + \frac{h\delta}{k_{air}} T_{ambient} - \frac{h\delta}{k_{air}} T_{plate}$$

$$= T_{plate} - \frac{\delta h}{k_{air}} \left(T_{plate} - T_{ambient} \right).$$

5.14. From the transmittance curve in Figure 5.11, estimate the values of *A*, *B* and *C*. The resist thickness used was 0.75 μm and the measurement was performed in the standard way. Assume a typical i-line resist.



From the curve, $T(0) = 0.32$, $T(\infty) = 0.88$, $dT/dE = 0.0022$ cm^2/mJ, and $T_{12} = 1 - [(1.69-1)/(1.69+1)]^2 = 0.934$.

$$A = \frac{1}{D} \ln\left[\frac{T(\infty)}{T(0)} \right] = \frac{1}{0.75} \ln\left[\frac{0.88}{0.32} \right] = 1.35 \ \mu\text{m}^{-1}$$

$$B = -\frac{1}{D} \ln\left[\frac{T(\infty)}{T_{12}} \right] = -\frac{1}{0.75} \ln\left[\frac{0.88}{0.934} \right] = 0.079 \ \mu\text{m}^{-1}$$

$$C = \frac{A + B}{AT(0)\left[1 - T(0)\right]T_{12}} \left. \frac{dT}{dE} \right|_{E=0} = \frac{1.35 + 0.08}{1.35(.32)\left[1 - 0.32\right](0.934)}(0.0022) = 0.011 \text{ cm}^2/\text{mJ}$$

Chapter 6

6.1. Equation (6.9) provides the (no diffusion or acid loss) kinetic solution for the blocking group concentration in a chemically amplified resist. Under the assumption that h is locally constant,

(a) Derive an expression for the relative bake time sensitivity of m (i.e., calculate $dm/d\ln t_{PEB}$).

(b) Derive an expression for the relative temperature sensitivity of m (i.e., calculate $dm/d\ln T$). From this, will a low activation energy resist or a high activation energy resist be more sensitive to temperature variations (for a given value of m)?

(c) Does the presence of base quencher change the bake time or temperature sensitivity of m (for a given value of m)?

$$m = e^{-K_{amp} t_{PEB} h}$$

a) $\dfrac{dm}{dt_{PEB}} = \left(-K_{amp} h\right) e^{-K_{amp} t_{PEB} h}$, $t_{PEB} \dfrac{dm}{dt_{PEB}} = \dfrac{dm}{d\ln(t_{PEB})} = \left(-K_{amp} t_{PEB} h\right) e^{-K_{amp} t_{PEB} h}$

$$\frac{dm}{d\ln(t_{PEB})} = m\ln(m)$$

b) $\dfrac{dm}{dT} = \left(-t_{PEB} h\right) e^{-K_{amp} t_{PEB} h} \dfrac{dK_{amp}}{dT}$, $T\dfrac{dm}{dT} = \dfrac{dm}{d\ln(T)} = \left(-t_{PEB} h\right) e^{-K_{amp} t_{PEB} h} \dfrac{dK_{amp}}{d\ln(T)}$

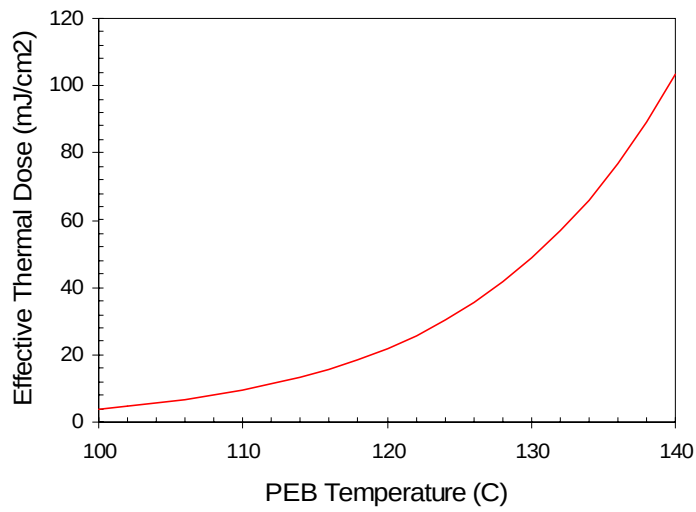
$$\frac{dm}{d\ln(T)} = \left(-K_{amp} t_{PEB} h\right) e^{-K_{amp} t_{PEB} h} \frac{d\ln(K_{amp})}{d\ln(T)} = m\ln(m)\frac{d\ln(K_{amp})}{d\ln(T)}$$

From the Arrhenius equation, $\ln(K_{amp}) = \ln(A_r) - E_a / RT$, $\dfrac{d\ln(K_{amp})}{dT} = \dfrac{E_a}{RT^2}$,

$\dfrac{d\ln(K_{amp})}{d\ln(T)} = \dfrac{E_a}{RT}$. Therefore, $\dfrac{dm}{d\ln(T)} = m\ln(m)\left(\dfrac{E_a}{RT}\right)$. A high activation energy resist will be more sensitive to relative temperature changes, for a given m and T.

c) With base quencher, $m = e^{-K_{amp} t_{PEB} (h - q_0)}$. Repeating all of the above calculations will give the same results. The time or temperature sensitivity does not change for a given value of m.

6.2. Plot the effective thermal dose versus temperature for a resist with an amplification reaction E_a = 25.0 kcal/mol and A_r = exp(28.0) s^{-1}, and an exposure rate constant C = 0.05 cm^2/mJ. Use a temperature range from 100 to 140 ºC and assume a PEB time of 60 s.



6.3. What is the impact of increasing the PAG loading (the initial concentration of PAG) on the effective thermal dose for a given PEB process?

Effective thermal dose = $K_{amp}t_{PEB}/C$, and K_{amp} = G_0k_4, where G_0 is the initial PAG loading. For a given PEB process (k_4 and t_{PEB} are fixed), increasing the PAG loading will increase the effective thermal dose.

6.4. Derive the simplified Fujita–Doolittle equation (6.15) from the full Fujita–Doolittle equation given in Chapter 5. What assumptions have to be made?

From Chapter 5, $D = D_0\, e^{-B(1/v_f - 1/v_g)}$, where $v_f = v_g + \alpha_2\left(T - T_g\right) + \beta\phi_s$. We want to derive $D_H = D_0 \exp\left(\dfrac{D_2(1-m)}{1+D_3(1-m)}\right)$. In this case, T is fixed and instead of solvent causing free volume we have deblocking. The fraction that is deblocked is 1-m. Thus, if we assume $T = T_g$,

$$v_f = v_g + \beta(1-m) \text{ and } -B(1/v_f - 1/v_g) = -B\left(\frac{1}{v_g + \beta(1-m)} - \frac{1}{v_g}\right) = -B\left(\frac{v_g - \left(v_g + \beta(1-m)\right)}{v_g\left(v_g + \beta(1-m)\right)}\right)$$

$$= \frac{B\beta(1-m)}{v_g\left(v_g + \beta(1-m)\right)} = \frac{B\beta}{v_g^{\,2}}\frac{(1-m)}{\left(1 + \dfrac{\beta}{v_g}(1-m)\right)}.$$ This gives the same result as equation (6.15) when

$$D_2 = \frac{B\beta}{v_g^{\,2}} \text{ and } D_3 = \frac{\beta}{v_g}.$$

6.5. Ignoring diffusion and acid loss mechanisms, derive an expression for the catalytic chain length (the number of deblocking events per acid molecule). At what dose will this catalytic chain length be at its maximum? What is the maximum value for the catalytic chain length?

Number of deblocking events = $(M_0 - M)$, Number of acids = H. Thus, the catalytic chain length = $\frac{(M_0 - M)}{H} = \frac{M_0}{G_0}\left(\frac{1-m}{h}\right) = \frac{M_0}{G_0}\left(\frac{1-e^{-K_{amp}t_{PEB}h}}{h}\right)$.

What dose (i.e., what value of h) will make this a maximum?

$$y = \frac{1-e^{-K_{amp}t_{PEB}h}}{h}, \quad \frac{dy}{dh} = 0 = \frac{h\left(K_{amp}t_{PEB}e^{-K_{amp}t_{PEB}h}\right) - \left(1-e^{-K_{amp}t_{PEB}h}\right)}{h^2}$$

$1 = \left(K_{amp}t_{PEB}h + 1\right)e^{-K_{amp}t_{PEB}h} = m\left[1+\ln(m)\right]$. This is true when m = 1, or h = 0. Thus, the maximum catalytic chain length occurs in the limit of zero dose. In this limit,

$\lim_{h\to 0}\left[\frac{1-e^{-K_{amp}t_{PEB}h}}{h}\right] = K_{amp}t_{PEB}$. Thus, the max catalytic chain length is $\frac{M_0}{G_0}K_{amp}t_{PEB}$.

6.6. One possible acid loss mechanism is called acid trapping, where acid is consumed by being trapped within the resist. Ignoring diffusion and all other acid loss mechanisms, this acid loss mechanism is governed by

$$\frac{\partial H}{\partial t_{PEB}} = -k_5 H$$

Under these conditions, derive an expression for the concentration of blocked sites at the end of the PEB.

Deblocking kinetics: $\frac{\partial M}{\partial t_{PEB}} = -k_4 M H$, so $\frac{\partial M}{M} = -k_4 H \partial t_{PEB}$ and $\int_{M_0}^{M}\frac{dM}{M} = -k_4\int_0^{t_{PEB}} H dt_{PEB}$.

But, we can determine $H(t)$ from the trapping kinetics: $H = H_0 e^{-k_5 t_{PEB}}$. So,

$$\ln\left(\frac{M}{M_0}\right) = -k_4\int_0^{t_{PEB}} H\partial t_{PEB} = -k_4 H_0\int_0^{t_{PEB}} e^{-k_5 t_{PEB}} dt_{PEB} = \frac{k_4}{k_5}H_0\left(e^{-k_5 t_{PEB}} - 1\right) = -\frac{k_4}{k_5}\left(H_0 - H\right)$$

$$M = M_0\exp\left[-\frac{k_4}{k_5}H_0\left(1-e^{-k_5 t_{PEB}}\right)\right] = M_0\exp\left[-K_{amp}t_{PEB}h_0\left(\frac{1-e^{-k_5 t_{PEB}}}{k_5 t_{PEB}}\right)\right].$$

6.7. For $x \gg \sigma_D$, derive a simplified 'large argument' approximation to the 1D reaction–diffusion point spread function.

$RDPSF(x) = 2\frac{e^{-x^2/2\sigma_D^2}}{\sqrt{2\pi}\sigma_D} - \frac{|x|}{\sigma_D^2}erfc\left(\frac{|x|}{\sqrt{2}\sigma_D}\right)$. For $x \gg \sigma_D$, the erfc has an asymptotic limit:

$\sqrt{\pi} z e^{z^2} erfc(z) \approx 1 - \frac{1}{2z^2}$. Thus, $RDPSF \approx 2\frac{e^{-x^2/2\sigma_D{}^2}}{\sqrt{2\pi}\sigma_D} - \frac{2}{\sqrt{2\pi}\sigma_D} e^{-x^2/2\sigma_D{}^2}\left(1 - \frac{\sigma_D{}^2}{x^2}\right)$

$$RDPSF \approx \sqrt{\frac{2}{\pi}}\frac{\sigma_D}{x^2} e^{-x^2/2\sigma_D{}^2}$$

6.8. Henry's law (first formulated in 1803) states that the amount of a gas which will dissolve into a solution is directly proportional to the partial pressure of that gas above the solution, and is generally found to be true only at low concentrations. When applied to a solid, the amount of a substance that is adsorbed onto the surface of the solid (the surface concentration) is proportional to the partial pressure of the substance in the atmosphere. The constant of proportionality, called the Henry's law constant, is dependent on both materials. Suppose ammonia is in the atmosphere at a concentration of 100 ppb (parts per billion), and that the Henry's law constant for ammonia in photoresist is 30 liter-atm/mol at room temperature. What is B_0, the surface concentration of ammonia?

$$B_0 = \left(\frac{mol}{30l - atm}\right)\left(100 \text{X} 10^{-9} atm\right) = 3.3 \text{ X } 10^{-9} \text{ mole/liter}$$

6.9. Derive an expression for the total amount of base contaminant found inside a resist, assuming that equation (6.22) applies and that σ_B « resist thickness. If one wanted to reduce this total amount of base contaminant by a factor of 2, how much would the post-exposure delay time have to be reduced (all other factors held constant)?

$B = B_0 \operatorname{erfc}\left(\frac{z}{\sqrt{2}\sigma_B}\right)$, Total contaminant = $\int_0^D B dz \approx \int_0^\infty B dz$ (D = resist thickness).

$$\int_0^\infty B dz = \int_0^\infty B_0 \operatorname{erfc}\left(\frac{z}{\sqrt{2}\sigma_B}\right) dz = B_0\left(z \operatorname{erfc}\left(\frac{z}{\sqrt{2}\sigma_B}\right) - \sqrt{\frac{2}{\pi}}\sigma_B e^{-z^2/2\sigma_B^2}\right)\Bigg|_0^\infty = B_0\sqrt{\frac{2}{\pi}}\sigma_B$$

Since diffusion length is proportional to the square root of delay time, to reduce the total base contaminant by a factor 2, one must reduce the delay time by a factor of 4.

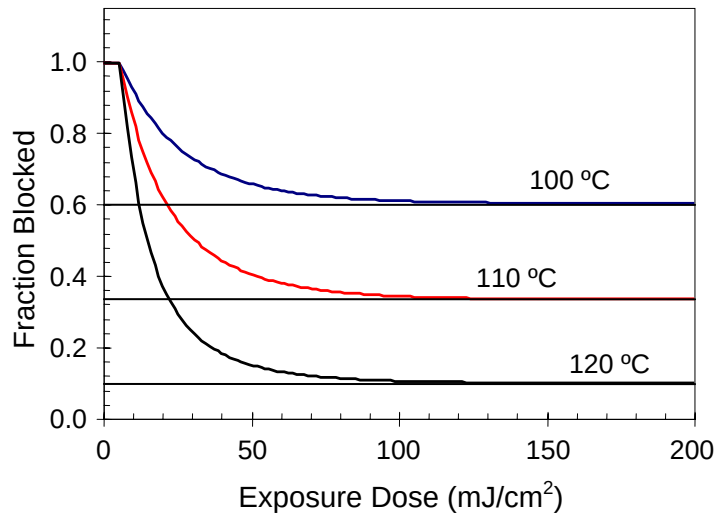
6.10. Why does the addition of base quencher reduce the sensitivity of the resist to airborne base contaminants?

The sensitivity to airborne base contaminants is reduced with the use of quencher for the same reason that the resist is less sensitive to acid diffusion: quencher results in a dramatic improvement in the latent image gradient near the resist feature edge (when processed at the dose to size). The error in edge position at the top of the resist (that is, the size of the T-top) is given approximately by

$\Delta x = \frac{acid\ loss\ due\ to\ airborne\ base}{dh/dx}$. Base quencher does not change the amount of acid loss, it increases the value of dh/dx.

6.11. For the deblocking kinetics data of Figure 6.7, assume a resist with an exposure rate constant C = 0.03 cm^2/mJ, and a PEB time of 60 s.

(a) If the resist shows no deblocking until a dose of 8 mJ/cm^2, what is the relative quencher loading for the resist?
(b) Estimate the amplification rate constant for each temperature shown.
(c) What is the activation energy for the amplification reaction?



a) $q_0 = h = 1 - e^{-CIt} = 1 - \exp[(-0.03)(8)] = 0.213$

b) At infinite dose, $h = 1$ and $m(\infty) = e^{-K_{amp}\, t_{PEB}\,(1-q_0)}$. For the three temperatures of the graphs,

At $T = 100$ °C, $m(\infty) = 0.6$ giving $K_{amp} = 0.0108$ s^{-1}

At $T = 110$ °C, $m(\infty) = 0.33$ giving $K_{amp} = 0.0234$ s^{-1}

At $T = 120$ °C, $m(\infty) = 0.1$ giving $K_{amp} = 0.0486$ s^{-1}

c) Fitting these three rate constants to a straight line in an Arrhenius plot gives E_a = 21.9 kcal/mol (91.6 kJ/mol).

6.12. Assuming that a certain amount of acid is required to achieve a desired lithographic effect (that is, assuming the mean concentration of photogenerated acid is fixed), how low can the mean number of photons go before photon shot noise exceeds the PAG loading shot noise? Assume, for example, that $\langle h \rangle$ = 0.3 and $\langle n_{0-PAG} \rangle / V$ = 0.05 /nm^3, and the region of interest is (10 nm)^3.

$$\sigma_h^2 = \frac{\langle h \rangle}{\langle n_{0-PAG} \rangle} + \frac{\left[\left(1-\langle h \rangle\right)\ln\left(1-\langle h \rangle\right)\right]^2}{\langle n \rangle}$$

. If the two components of variance are the same,

$$\frac{\langle h \rangle}{\langle n_{0-PAG} \rangle} = \frac{\left[\left(1-\langle h \rangle\right)\ln\left(1-\langle h \rangle\right)\right]^2}{\langle n \rangle} \text{ and } \langle n \rangle = \langle n_{0-PAG} \rangle \frac{\left[\left(1-\langle h \rangle\right)\ln\left(1-\langle h \rangle\right)\right]^2}{\langle h \rangle}$$

. For $h = 0.3$,

$$\frac{\left[\left(1-\langle h \rangle\right)\ln\left(1-\langle h \rangle\right)\right]^2}{\langle h \rangle} = 0.208, \text{ so that } \langle n \rangle = 0.208(0.05)(10)^3 = 10.4 \text{ photons.}$$

6.13. Suppose for a given process that $\langle h \rangle$ = 0.3 and $\langle m \rangle$ = 0.5. Further, assume that photon shot noise can be neglected and that $\langle n_{0-blocked} \rangle / \langle n_{0-PAG} \rangle$ = 30. Considering the total uncertainty of the final blocked polymer concentration for a 1D problem, how large must σ_D/a be in order to make the contribution of acid uncertainty equal to the Poisson uncertainty of the deblocking reaction?

Ignoring photon shot noise, $\sigma_m^2 = \dfrac{\langle m \rangle}{\langle n_{0-blocked} \rangle} + \left(\langle m \rangle \ln \langle m \rangle\right)^2 \left(\dfrac{a}{\sigma_D}\right)^p \left(\dfrac{1}{\langle n_{0-PAG} \rangle \langle h \rangle}\right)$.

Making the two components of variance equal,

$$\frac{\langle m \rangle}{\langle n_{0-blocked} \rangle} = \left(\langle m \rangle \ln \langle m \rangle\right)^2 \left(\frac{a}{\sigma_D}\right)^p \left(\frac{1}{\langle n_{0-PAG} \rangle \langle h \rangle}\right) \quad \text{or} \quad \frac{\langle n_{0-PAG} \rangle \langle h \rangle}{\langle m \rangle \left(\ln \langle m \rangle\right)^2 \langle n_{0-blocked} \rangle} = \left(\frac{a}{\sigma_D}\right)^p .$$

$$\frac{\sigma_D}{a} = \frac{\langle m \rangle \left(\ln \langle m \rangle\right)^2 \langle n_{0-blocked} \rangle}{\langle n_{0-PAG} \rangle \langle h \rangle} = \frac{0.5\left(\ln(0.5)\right)^2 (30)}{0.3} = 24.$$

6.14. As equation (6.110) shows, increasing acid diffusion during PEB reduces the stochastic uncertainty caused by acid concentration fluctuations. Is there a limit as to how much this uncertainty can be reduced (that is, is there a practical maximum value for the acid diffusion length)? Explain.

Yes. Too much diffusion will diffuse away the features being printed! In practice, the diffusion length must be small compared to the minimum feature size.

6.15. Qualitatively, how will the presence of base quencher affect the variance of final blocked polymer concentration?

Good question! I'm not sure that I know the answer, but it will be interesting to see what the students think!

Chapter 7

7.1. For the original kinetic model of equation (7.10), what is the value of the development rate at $m = m_{th}$? For $m_{th} > 0$, what is the limit of this value as n becomes large?

$$r = r_{max}\frac{(a+1)(1-m)^n}{a+(1-m)^n} + r_{min}, \quad a = \frac{(n+1)}{(n-1)}\left(1-m_{th}\right)^n$$

At $m = m_{th}$, $r = r_{max}\dfrac{(a+1)(1-m_{th})^n}{a+(1-m_{th})^n} + r_{min} = r_{max}\dfrac{a+1}{\dfrac{a}{(1-m_{th})^n}+1} + r_{min}$

$$= r_{max}\frac{a+1}{\dfrac{(n+1)}{(n-1)}+1} + r_{min} = r_{max}\frac{(a+1)(n-1)}{2n} + r_{min} = r_{max}\frac{(n+1)(1-m_{th})^n + (n-1)}{2n} + r_{min}$$

As n gets very large, $(1-m_{th})^n$ goes to zero for $m_{th} > 0$. Thus, for this case,

$$r(m_{th}) \approx r_{max}\frac{n-1}{2n} + r_{min} \approx \frac{r_{max}}{2} + r_{min}$$

7.2. Derive equation (7.9).

$$r = r_{max}\frac{(a+1)(1-m)^n}{a+(1-m)^n} + r_{min}, \text{ show that } \frac{d^2r}{dm^2} = 0 \quad \text{at} \quad m = m_{th} \text{ gives } a = \frac{(n+1)}{(n-1)}(1-m_{th})^n$$

$$\frac{dr}{dm} = r_{max}\frac{-\left[a+(1-m)^n\right]n(a+1)(1-m)^{n-1} + (a+1)(1-m)^n n(1-m)^{n-1}}{\left[a+(1-m)^n\right]^2} = -anr_{max}\frac{(a+1)(1-m)^{n-1}}{\left[a+(1-m)^n\right]^2}$$

$$= \frac{-an\left(r-r_{min}\right)}{(1-m)\left[a+(1-m)^n\right]}$$

$$\frac{d^2r}{dm^2} = 0 = -an\frac{(1-m)\left[a+(1-m)^n\right]\left(dr/dm\right) + \left(r-r_{min}\right)\left[n(1-m)^n + a + (1-m)^n\right]}{(1-m)^2\left[a+(1-m)^n\right]^2}$$

$$0 = (1-m)\left[a+(1-m)^n\right]\left(dr/dm\right) + \left(r-r_{min}\right)\left[n(1-m)^n + a + (1-m)^n\right]$$

$$0 = -an\left(r-r_{min}\right) + \left(r-r_{min}\right)\left[n(1-m)^n + a + (1-m)^n\right]$$

$$0 = -an\left(r - r_{min}\right) + \left(r - r_{min}\right)\left[(n+1)(1-m)^n + a\right]$$

$$a = \frac{(n+1)}{n-1}(1-m)^n$$

7.3. Using the same approach that was used for the original kinetic development model, add the effect of developer diffusion to the surface of the resist to the enhanced kinetic model. In other words, let equation (7.12) represent the surface reaction rate and assume that r_{resin} is proportional to D_S.

$r_R = r_{resin}\dfrac{1+k_{enh}(1-m)^n}{1+k_{inh}\,(m)^l} = r_R = k_R D_S \dfrac{1+k_{enh}(1-m)^n}{1+k_{inh}\,(m)^l}$ and $r_D = k_D\left(D - D_S\right)$. Setting these two rates equal to each other and solving for D_S,

$$k_D\left(D - D_S\right) = k_R D_S \frac{1+k_{enh}(1-m)^n}{1+k_{inh}\,(m)^l}, \quad k_D D = D_S\left[k_R \frac{1+k_{enh}(1-m)^n}{1+k_{inh}\,(m)^l} + k_D\right],$$

$$D_S = \frac{k_D D}{k_R \dfrac{1+k_{enh}(1-m)^n}{1+k_{inh}\,(m)^l} + k_D} \quad \text{and} \quad r = \frac{k_R k_D D}{k_R \dfrac{1+k_{enh}(1-m)^n}{1+k_{inh}\,(m)^l} + k_D}\,\frac{1+k_{enh}(1-m)^n}{1+k_{inh}\,(m)^l}$$

$$r = \frac{k_R k_D D\left(1+k_{enh}(1-m)^n\right)}{k_R\left(1+k_{enh}(1-m)^n\right) + k_D\left(1+k_{inh}\,(m)^l\right)}.$$

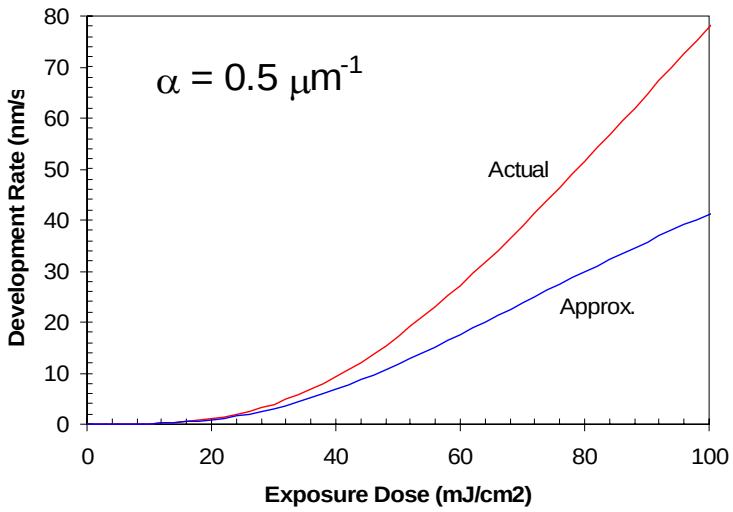
7.4. Consider a conventional positive resist that obeys the reaction-controlled version of the original kinetic development model of equation (7.11). Further assume no bleaching, but absorption so that the inhibitor concentration m can be related to incident dose E and depth into resist z by

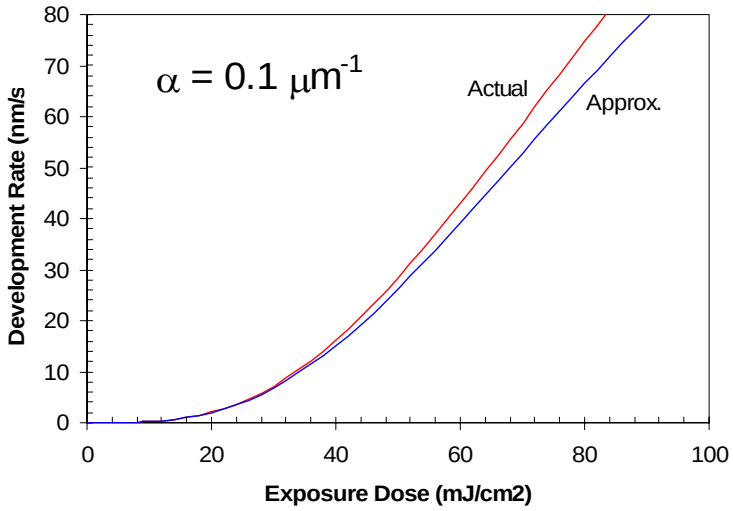
$$m = e^{-CE\,I_r(z)} \qquad \text{where} \qquad I_r(z) = e^{-\alpha z}$$

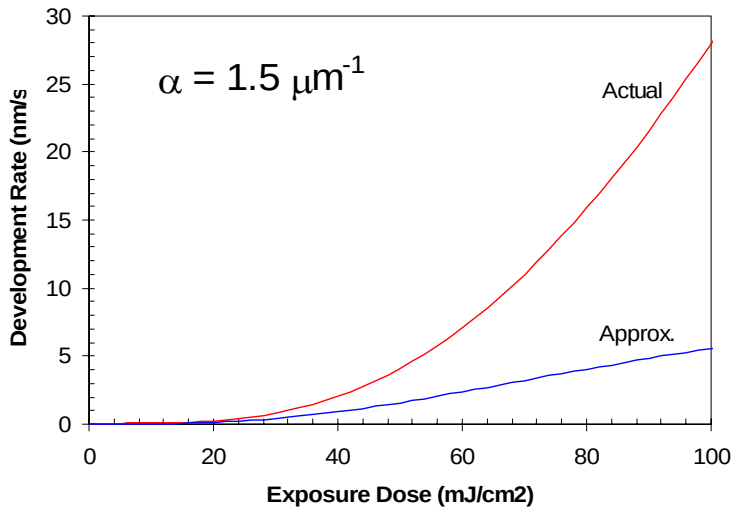
Using a spreadsheet or similar program, plot development rate versus dose (up to 100 mJ/cm^2) for r_{max} = 200 nm/s, r_{min} = 0.05 nm/s, n = 4, C = 0.02 cm^2/mJ, α = 0.5 μm^{-1}, and a depth into the resist of 500 nm. Compare these results to the approximate equation (7.19):

$$r(z) \approx r(z=0)\,e^{-\alpha n z}$$

Repeat for α = 0.1 μm^{-1} and α = 1.5 μm^{-1}. What can you conclude about the accuracy of the approximate expression?


α = 0.5 μm-1
Actual
Approx.
Development Rate (nm/s
Exposure Dose (mJ/cm2)
0
10
20
30
40
50
60
70
80
0
20
40
60
80
100


α = 0.1 μm-1
Actual
Approx.
Development Rate (nm/s
Exposure Dose (mJ/cm2)
0
10
20
30
40
50
60
70
80
0
20
40
60
80
100



This approximation is only reasonably accurate at low doses, but is much more accurate for low α than high α.

7.5. Consider a conventional positive resist that obeys the reaction-controlled version of the original kinetic development model of equation (7.11). Further, assume no bleaching so that the inhibitor concentration m can be related to dose E by

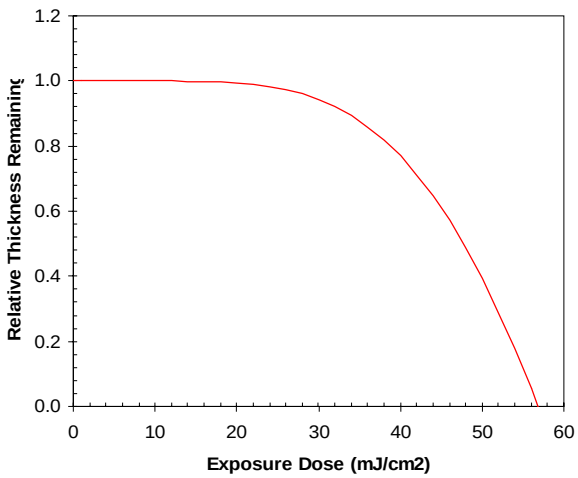
$$m = e^{-CE}$$

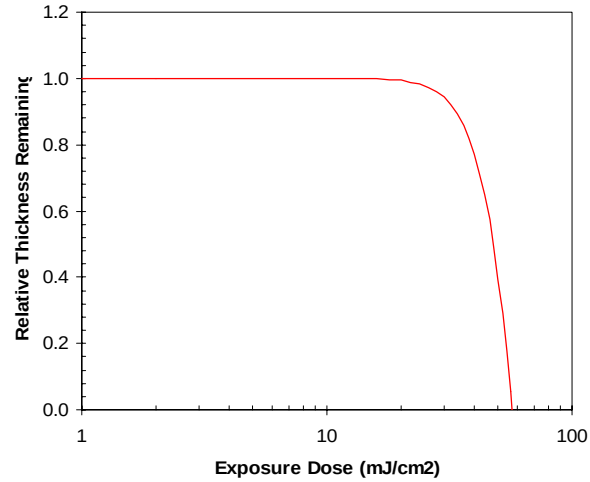
Assuming that the minimum development rate r_{min} can be ignored over the dose range of interest and ignoring any variation in development rate with depth into the resist, derive an expression for the resist characteristic curve (such as that pictured in Figure 7.11). Plot the resulting equation for r_{max} = 200 nm/s, n = 7, C = 0.02 cm^2/mJ, an initial resist thickness of 800 nm, and a development time of 60 s. What is the value of the dose-to-clear for this case?

$r = r_{max}\left(1 - e^{-CE}\right)^n$ and $t_{dev} = \int_0^z \frac{dz'}{r(z')}$. Since development rate doesn't vary with z,

$z = rt_{dev} = r_{max}t_{dev}\left(1 - e^{-CE}\right)^n$ and the relative thickness remaining is

$$T_r = \frac{D - z}{D} = 1 - \frac{r_{max}t_{dev}}{D}\left(1 - e^{-CE}\right)^n$$





Dose-to-clear is about 56.8 mJ/cm^2. Note that the dose-to-clear can be derived from the above equation by setting $T_r = 0$:

$$\frac{r_{max}t_{dev}}{D}\left(1-e^{-CE_0}\right)^n = 1, \quad E_0 = -\frac{1}{C}\ln\left(1-\sqrt[n]{\frac{D}{r_{max}t_{dev}}}\right)$$

7.6. Given a conventionally measured resist characteristic curve, estimate how a small change in resist thickness would change the dose-to-clear.

From the definition of measured contrast: $\gamma_m = \frac{1}{D}\frac{dz}{d\ln E}\bigg|_{E=E_0}$. If the resist is a little thicker, the depth developed into the resist z will have to go a little further to hit bottom, which will require a little higher dose. Since $z = D$ at the bottom,

$\gamma_m = \frac{d\ln z}{d\ln E}\bigg|_{E=E_0} = \frac{\Delta z}{z}\Big/\frac{\Delta E}{E}\bigg|_{E=E_0} = \frac{\Delta D}{D}\Big/\frac{\Delta E_0}{E_0}$. Assuming γ_m is constant over this small range of resist thicknesses, $\frac{\Delta E_0}{E_0} = \frac{1}{\gamma_m}\frac{\Delta D}{D}$. QED.

As a specific example, consider the solution from the previous problem:

$E_0 = -\frac{1}{C}\ln\left(1-\sqrt[n]{\frac{D}{r_{max}t_{dev}}}\right)$. Taking the derivative with respect to D,

$\frac{dE_0}{dD} = \frac{1}{nC}\frac{1}{1-\sqrt[n]{\frac{D}{r_{max}t_{dev}}}}\left(\frac{D}{r_{max}t_{dev}}\right)^{\frac{1}{n}-1}\left(\frac{1}{r_{max}t_{dev}}\right) = \frac{1}{nC}\left(\frac{1-e^{-cE_0}}{e^{-cE_0}}\right)\left(\frac{1}{D}\right) = \frac{e^{cE_0}-1}{nCD}$. But, from the same solution we can calculate γ_m: $\gamma_m = \frac{nCE_0}{e^{cE_0}-1}$. Thus, $\frac{dE_0}{dD} = \frac{E_0}{\gamma_m D}$. For small changes in D, we can assume γ_m is constant and $\frac{\Delta E_0}{E_0} = \frac{1}{\gamma_m}\frac{\Delta D}{D}$.

7.7. Derive equation (7.32).

Given $\gamma_m = \frac{r_B}{D}\gamma_{th}\int_0^D \frac{dz}{r_B\left[1-(1-r_0)e^{-z/\delta}\right]}$, derive $\gamma_m = \gamma_{th}\left[1-\frac{\delta}{D}\ln(r_0)\right]$. Obviously, the r_B terms cancel, so that we need to solve the integral $\gamma_m = \frac{\gamma_{th}}{D}\int_0^D \frac{dz}{\left[1-(1-r_0)e^{-z/\delta}\right]}$. Letting $y = e^{z/\delta}$, the integral becomes $\frac{\gamma_m}{\gamma_{th}} = \frac{\delta}{D}\int_1^{e^{D/\delta}} \frac{dy}{y-(1-r_0)} = \frac{\delta}{D}\ln(y-(1-r_0))\Big|_1^{e^{D/\delta}} = \frac{\delta}{D}\ln\left(e^{D/\delta}-(1-r_0)\right) - \frac{\delta}{D}\ln(r_0)$

Assuming $D \gg \delta$, $\ln\left(e^{D/\delta}-(1-r_0)\right) \approx \frac{D}{\delta}$ and $\gamma_m = \gamma_{th}\left[1-\frac{\delta}{D}\ln(r_0)\right]$.

7.8. Derive equation (7.63).

The slope of the development path is $z' = \frac{r(x)}{\sqrt{r^2(x_0)-r^2(x)}}$. The resist profile is perpendicular to this path so that its slope is -1/z'. Thus, the sidewall angle is given by the tangent to the slope:

$$\tan\theta = \frac{1}{z'} = \frac{\sqrt{r^2(x_0)-r^2(x)}}{r(x)} = \frac{\sqrt{1-\frac{r^2(x)}{r^2(x_0)}}}{\frac{r(x)}{r(x_0)}} = \frac{\sin\theta}{\cos\theta} = \frac{\sqrt{1-\cos^2\theta}}{\cos\theta}.$$ Thus, $\cos\theta = \frac{r(x)}{r(x_0)}$.

7.9. For the Lumped Parameter Model, consider the case where only absorption causes a variation of development rate with *z*. In the limit of very large contrast, what is D_{eff} for the two cases of $\alpha = 0$ and $\alpha > 0$?

$$D_{eff} = \frac{1}{\alpha\gamma}\left(1-e^{-\alpha\gamma D}\right).$$

$$\alpha > 0: \quad \lim_{\gamma\to\infty} \frac{1}{\alpha\gamma}\left(1 - e^{-\alpha\gamma D}\right) = \lim_{\gamma\to\infty} \frac{1}{\alpha\gamma} = 0$$

$$\alpha = 0: \quad \lim_{\alpha\to 0} \frac{1}{\alpha\gamma}\left(1 - e^{-\alpha\gamma D}\right) = D \text{ for any value of } \gamma.$$

7.10. The assumption of a segmented development path can lead to a calculation of the resulting CD of a space for a given development rate function $r(x,z)$ and a given development time. How will this CD compare to the CD calculated by rigorously solving the Euler–Lagrange equation for the same conditions?

The rigorously calculated path is, by definition, the fastest path. All other paths, including the assumed segmented path, will arrive at the resist edge more slowly. Thus, the predicted space width will always be smaller for the segmented path than fro the rigorous path.

7.11. Use the Lumped Parameter Model to calculate CD as a function of dose for the case of a space that has a constant log-slope aerial image:

$$I(x) = I_0 e^{-s|x|}$$

where s is positive. Also assume that $x_0 = 0$ and that there is no z-dependence to the development rate. Compare this result to equation (7.76).

$$\left(\frac{E(x)}{E(0)}\right)^{\gamma} = 1 + \frac{1}{D}\int_{x_0}^{x}\left(\frac{I(x')}{I(x_0)}\right)^{-\gamma} dx' = 1 + \frac{1}{D}\int_{0}^{x} e^{s\gamma x'}\, dx' = 1 + \frac{e^{s\gamma x} - 1}{s\gamma D}.$$ Solving for x,

$$x = \frac{1}{s\gamma}\ln\left(s\gamma D\left(\frac{E(x)}{E(0)}\right)^{\gamma} - s\gamma D + 1\right),$$ where $CD = 2x$.

By comparison, equation (7.76) gives the full Euler-Lagrange solution for the same development rate expression with $a = s\gamma$: $x = \frac{1}{a}\ln\left(\sqrt{[ar(0)t]^2 + 1}\right)$. But since $r(0)t$ for the LPM is given by $D\left(\frac{E(x)}{E(0)}\right)^{\gamma}$ from equation (7.86), the LPM solution becomes $x_{LPM} = \frac{1}{a}\ln\left(ar(0)t - aD + 1\right)$. Note that these two solutions become the same if $r(0)t \gg D$ and $ar(0)t \gg 1$. Assuming this is true and expanding both solutions as Taylor series:

$$x_{E-L} \approx \frac{1}{a}\ln\left(ar(0)t\right) + \frac{1}{2a^2 r(0)t}$$

$$x_{LPM} \approx \frac{1}{a}\ln\left(ar(0)t\right) + \frac{1 - aD}{a^2 r(0)t} \quad \text{and} \quad x_{LPM} \approx x_{E-L} - \frac{aD - 0.5}{a^2 r(0)t}$$

7.12. For the Lumped Parameter Model, consider the case where only absorption causes a variation of development rate with *z*. Derive an expression that compares the value of resist contrast as measured with a standard photoresist Hurter–Driffield (characteristic) curve to the theoretical (LPM) contrast.

$$t_{dev} = \int_0^D \frac{dz}{r(z)} = \left(\frac{E_0}{E}\right)^\gamma \frac{1}{r_0}\int_0^D \frac{dz}{\left(I_r(z)\right)^\gamma} = \left(\frac{E_0}{E}\right)^\gamma \frac{1}{r_0}\int_0^z e^{\alpha\gamma z}dz = \left(\frac{E_0}{E}\right)^\gamma \frac{e^{\alpha\gamma z}-1}{r_0\alpha\gamma}.$$ Solving for *z*,

$$z = \frac{1}{\alpha\gamma}\ln\left(\alpha\gamma r_0\left(\frac{E}{E_0}\right)^\gamma t_{dev} + 1\right).$$

Using the definition of measured contrast, $\gamma_m = \pm\frac{1}{D}\frac{dz}{d\ln E}\bigg|_{E=E_0}$, gives $\gamma_m = \frac{1-e^{-\alpha\gamma D}}{\alpha D}$.

Note that a Taylor series expansion of the exponential gives $\gamma_m = \gamma - \frac{\alpha D\gamma^2}{2} + \ldots$

Chapter 8

8.1. A given 90 nm lithography process level exhibits a 0.1 ppm (parts per million) wafer magnification error due to wafer expansion. For a 300 mm wafer, what is the maximum overlay error due to this term if it goes uncorrected? Is the magnitude of this error significant (assume the overlay specifications is 1/3 of the technology node feature size)?

Wafer mag error at the edge of the wafer (assuming that the center of the wafer is properly aligned) will result in an overlay error of 150 mm X 0.1×10^{-6} = 15 nm. (Note that the opposite edge of the wafer would have an error of -15 nm.) Since the overlay spec is about 90/3 = 30 nm, this uncorrected wafer mag error would amount to one half of the overlay budget.

8.2. Silicon has a coefficient of linear expansion of about 3×10^{-6}/K. Assume that 3 nm of overlay error due to lack of wafer temperature control is acceptable. How well must the temperature of the wafer be controlled for a 200 mm wafer? For a 300mm wafer?

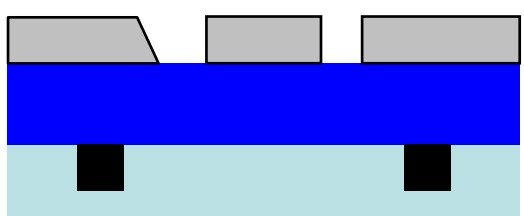
Assuming that the center of the wafer is properly aligned, at the edge of the 200 mm wafer, the expansion that could be tolerated would be $(100mm)\left(3\times10^{-6}/K\right)\Delta T = 3nm$, or ΔT = 10 mK. For a 300 mm wafer, the control must be better than 6.7 mK.

8.3. What are the maximum x- and y-overlay errors due to a 0.3 arc-second reticle rotation error for a 24 mm X 28 mm field?

$dx = -\phi y$, $dy = -\phi x$. A 0.3 arc-second rotation is 0.3/3600 = 0.0000833 degrees = 1.45×10^{-6} radians. Assuming that the center of the field is kept in alignment, the max overlay errors become

$$|dx| = 0.00000145(14mm) = 20.4nm, \quad |dy| = 0.00000145(12mm) = 17.5nm$$

8.4. It is possible that the overlay targets, when printed on the wafer, may be asymmetric, as shown below for a frame-in-frame target. What will be the impact of such asymmetry on overlay measurement?



The measured overlay will depend on where in *z* the measurement is made. When measuring at the bottom of the resist, the inner box looks like it is further to the right compared to making a

measurement at the top of the resist. For an optical image-based measurement, the measured overlay will depend on the focus of the imaging tool.

8.5. Derive equations (8.42) and (8.43).

Derive the dose-dependence of the log-CD versus log-dose slope: for the simplest threshold model: $\frac{d\ln CD}{d\ln E}=\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\left(\frac{E_1}{E}\right)\left(\frac{CD_1}{CD}\right)$ and for the Gaussian image: $\frac{d\ln CD}{d\ln E}=\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\left(\frac{CD_1}{CD}\right)^2$ (for a space).

Start with the constant slope model and take its derivative with respect to dose:

$$\frac{CD-CD_1}{CD_1}=\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\left(1-\frac{E_1}{E}\right),\quad \frac{1}{CD_1}\frac{dCD}{dE}=\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\left(\frac{E_1}{E^2}\right)$$

$$\frac{CD}{CD_1}\frac{d\ln CD}{d\ln E}=\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\left(\frac{E_1}{E}\right),\quad \frac{d\ln CD}{d\ln E}=\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\left(\frac{E_1}{E}\right)\left(\frac{CD_1}{CD}\right)$$

Next, take the more exact Gaussian image model and take its derivative with respect to dose:

$$\frac{CD}{CD_1}=\sqrt{1+2\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\ln\left(\frac{E}{E_1}\right)},$$

$$\frac{1}{CD_1}\frac{dCD}{dE}=\frac{1}{2}\left[1+2\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\ln\left(\frac{E}{E_1}\right)\right]^{-\frac{1}{2}}\left(\frac{2}{E}\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\right)$$

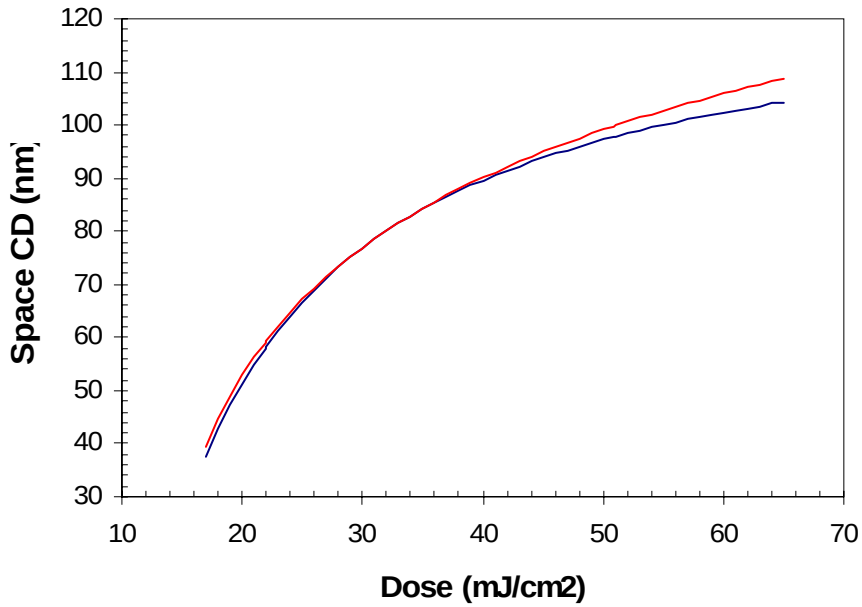
$$\frac{1}{CD_1}\frac{dCD}{d\ln E}=\left[1+2\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\ln\left(\frac{E}{E_1}\right)\right]^{-\frac{1}{2}}\left(\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\right)=\left(\frac{CD_1}{CD}\right)\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}$$

$\frac{d\ln CD}{d\ln E}=\left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}\left(\frac{CD_1}{CD}\right)^2$. The derivation for the line is very similar.

8.6. Equations (8.31) and (8.35) both predict CD versus dose, with the later equation a bit more accurate. Plot CD versus dose for these two models on one graph for the case of

$$CD_1 = 80\text{ nm},\ E_1 = 32\text{ mJ/cm2},\ \text{and}\ \left.\frac{d\ln CD}{d\ln E}\right|_{E=E_1}=0.6$$

At what dose values do the two models disagree with each other in predicted CD by 2%?

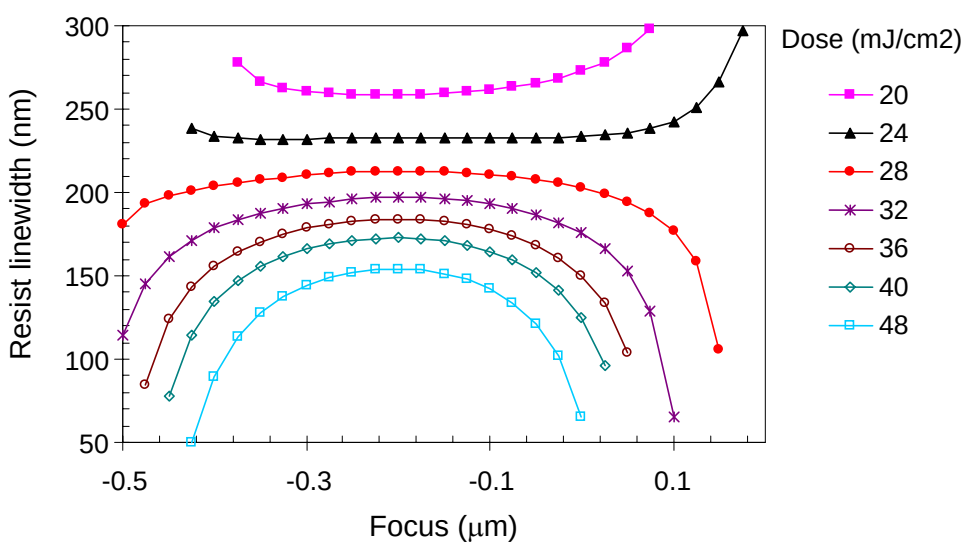


The two models disagree in CD by 2% at the doses of 21.93 mJ/cm^2 (CD = 59.2 nm) and 50.88 mJ/cm^2 (CD = 99.8 nm).

8.7. For the Bossung plot given below,

a) Approximately where is best focus? Why?

b) What is the approximate isofocal bias (numerical value) for a target CD of 180 nm?



a) Best focus, estimated as the peak of the nearly parabolic CD through focus curves, is at about -0.2 μm. At this focus, CD is least sensitive to focus errors. Also, at this focus the different dose curves are closest together, indicating maximum exposure latitude.

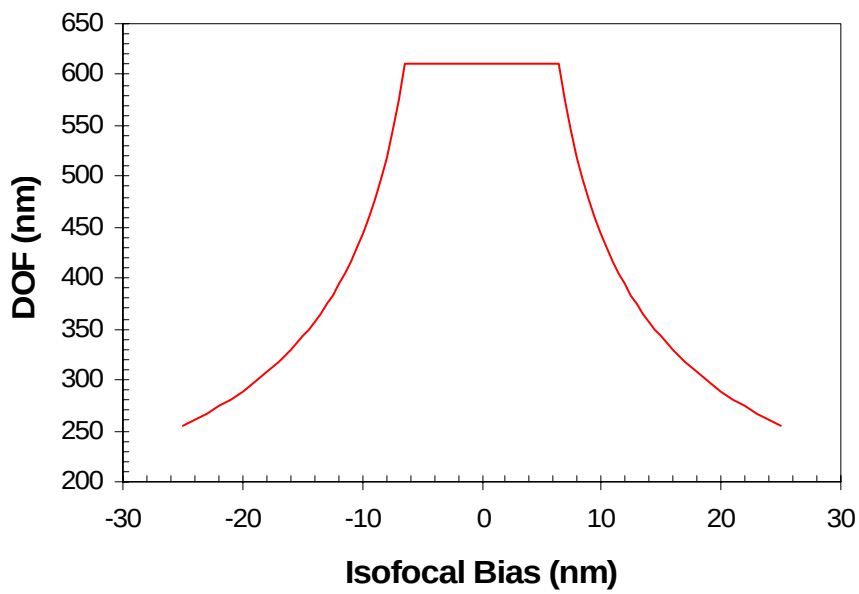
b) I would estimate the isofocal dose at slightly more than 24 mJ/cm^2 (maybe 25 would be a good estimate) – the point where CD through focus is the flattest over the longest range of focus. This puts the isofocal CD at about 230 nm, for an isofocal bias of 50 nm.

8.8. Using equations (8.50) – (8.52), plot the depth of focus of a 65 nm line as a function of isofocal bias: let $CD_{spec} = 0.2$, $El_{spec} = 0.08$, $s_{nom} = 0.75$, and $\Delta = 0.2$ µm.

For isofocal bias within the CD spec: $DOF = 2\Delta\sqrt{\dfrac{CD_{spec}}{|s_{nom}|EL_{spec}} - 1}$

For isofocal bias greater than CD spec range: $DOF \approx \dfrac{1}{K}2\Delta\sqrt{\dfrac{CD_{spec}}{|s_{nom}|EL_{spec}} - 1}$ where

$K = \sqrt{\dfrac{CD_{rel-iso-limit}}{|s_{nom}|EL_{spec}} + 1}$ and $CD_{rel-iso-limit} = \dfrac{|CD_1 - CD_{limit}|}{CD_{nom}}$



8.9. Use the general expression (8.75) to derive the image MEEF for a coherent equal line/space image that uses only the 0th and ±1st diffraction orders (that is, derive equation (8.68)).

Use $image\ MEEF = \dfrac{2}{NILS}\dfrac{\partial \ln I}{\partial \ln w}\Big|_{x=CD/2}$ to derive the coherent image MEEF:

$$image\ MEEF = \frac{2+\cos(2\pi w/p)}{1-\cos(2\pi w/p)} \quad \text{for} \quad I(x) = \left[\frac{w}{p} + \frac{2\sin(\pi w/p)}{\pi}\cos(2\pi x/p)\right]^2.$$

$$\text{NILS:} \quad -\frac{dI}{dx} = 2\left[\frac{w}{p} + \frac{2\sin(\pi w/p)}{\pi}\cos(2\pi x/p)\right]\frac{2\sin(\pi w/p)}{\pi}\sin(2\pi x/p)\frac{2\pi}{p}$$

$$-\left.\frac{d\ln I}{dx}\right|_{x=w/2} = \frac{8}{p}\left[\frac{w}{p}+\frac{2\sin(\pi w/p)}{\pi}\cos(\pi w/p)\right]^{-1}\sin^2(\pi w/p)$$

$$\frac{2}{NILS} = \frac{p}{4w\sin^2(\pi w/p)}\left[\frac{w}{p}+\frac{2\sin(\pi w/p)}{\pi}\cos(\pi w/p)\right]$$

$$\frac{\partial I}{\partial w} = 2\left[\frac{w}{p}+\frac{2\sin(\pi w/p)}{\pi}\cos(2\pi x/p)\right]\left[\frac{1}{p}+\frac{2\cos(\pi w/p)}{p}\cos(2\pi x/p)\right]$$

$$\left.\frac{\partial \ln I}{\partial \ln w}\right|_{x=CD/2} = \frac{2w}{p}\left[\frac{w}{p}+\frac{2\sin(\pi w/p)}{\pi}\cos(\pi w/p)\right]^{-1}\left[1+2\cos^2(\pi w/p)\right]$$

$$image\ MEEF = \frac{2}{NILS}\left.\frac{\partial \ln I}{\partial \ln w}\right|_{x=CD/2} = \frac{\left[1+2\cos^2(\pi w/p)\right]}{2\sin^2(\pi w/p)}$$

Using the double angle formulas, $image\ MEEF = \dfrac{\left[2+\cos(2\pi w/p)\right]}{1-\cos(\pi w/p)}$. QED

8.10. For the two-line pattern collapse problem, what linewidth to spacewidth ratio (for a fixed pitch) minimizes the tendency for pattern collapse under the following conditions:

a) the assume that the water-resist contact angle is small enough that the $\sin\theta$ term in equation (8.83) can be ignored.

b) the assume that the water-resist contact angle is large enough that the $\cos\theta$ term in equation (8.83) can be ignored.

a) $\dfrac{E}{\sigma} \le \dfrac{8}{w_s}A_l^3\left[3A_s\cos\theta+\sin\theta\right] \approx \dfrac{24}{w_s}A_l^3 A_s\cos\theta = \dfrac{24H^4}{w_s{}^2 w_l{}^3}\cos\theta$. For a given H and θ, we want to maximize $w_s{}^2 w_l{}^3$. Let $p = w_s + w_l$, so we want to maximize $(p-w_l)^2 w_l{}^3$. Taking the derivative and setting it equal to zero,

$$0 = -2(p-w_l)w_l{}^3 + 3(p-w_l)^2 w_l{}^2 \quad \text{or} \quad 0 = -2w_l + 3(p-w_l), \text{ giving } w_l = \frac{3p}{5} \text{ and } \frac{w_l}{w_s} = 1.5.$$

b) $\dfrac{E}{\sigma} \le \dfrac{8}{w_s}A_l^3\left[3A_s\cos\theta+\sin\theta\right] \approx \dfrac{8}{w_s}A_l^3\sin\theta = \dfrac{8H^3}{w_s w_l{}^3}\sin\theta$. For a given H and θ, we want to maximize $w_s w_l{}^3$. Let $p = w_s + w_l$, so we want to maximize $(p-w_l)w_l{}^3$. Taking the derivative and setting it equal to zero,

$$0 = -w_l{}^3 + 3(p-w_l)w_l{}^2 \quad \text{or} \quad 0 = -w_l + 3(p-w_l), \text{ giving } w_l = \frac{3p}{4} \text{ and } \frac{w_l}{w_s} = 3.$$

Chapter 9

9.1. Using results from Chapter 3, show that the aerial image NILS falls off approximately quadratically with defocus.

Consider images with defocus from Table 3.2.

three-beam TE coherent: $NILS = 8\cos(\Delta\Phi)$, $\Delta\Phi = 2\pi n\delta\left(1-\sqrt{1-(\lambda/np)^2}\right)/\lambda$. For small defocus,

$\cos(\Delta\Phi) \approx 1 - \dfrac{\Delta\Phi^2}{2}$ so that $NILS \approx 8 - 4\Delta\Phi^2 = 8 - 16\pi^2 n^2 \delta^2 \left(1-\sqrt{1-(\lambda/np)^2}\right)^2 / \lambda^2$

Or, $NILS \approx NILS_0\left(1-\left(\dfrac{\delta}{\Delta}\right)^2\right)$, where $\Delta = \dfrac{\lambda}{\sqrt{2}\pi n\left(1-\sqrt{1-(\lambda/np)^2}\right)}$ and $NILS_0 = 8$.

Partially coherent three-beam out of focus: $NILS = \dfrac{8\cos(\Delta\Phi)\dfrac{2J_1(a)}{a}}{1+\dfrac{8}{\pi^2}\left(1-\dfrac{J_1(2a)}{a}\right)}$. For small defocus,

$\dfrac{2J_1(a)}{a} \approx 1 - \dfrac{a^2}{8}$, $a = 2\pi\,\delta\,\sigma\,NA/p$. Thus, keeping out to only the second order terms,

$8\cos(\Delta\Phi)\dfrac{2J_1(a)}{a} \approx 8\left(1-\dfrac{a^2}{8}-\dfrac{\Delta\Phi^2}{2}\right)$ and $\dfrac{1}{1+\dfrac{8}{\pi^2}\left(1-\dfrac{J_1(2a)}{a}\right)} \approx \dfrac{1}{1+\dfrac{a^2}{\pi^2}} \approx 1-\dfrac{a^2}{\pi^2}$

$NILS \approx 8\left(1-\dfrac{a^2}{8}-\dfrac{\Delta\Phi^2}{2}-\dfrac{a^2}{\pi^2}\right)$. So, $NILS \approx NILS_0\left(1-\left(\dfrac{\delta}{\Delta}\right)^2\right)$ where $NILS_0 = 8$ and

$$\Delta = \frac{\lambda}{n\sqrt{\left(\dfrac{\pi^2}{2}+4\right)\sigma^2 NA^2\lambda^2/p^2 + 2\pi^2\left(1-\sqrt{1-(\lambda/np)^2}\right)}}$$

9.2. Consider the three-beam aerial image of lines and spaces:

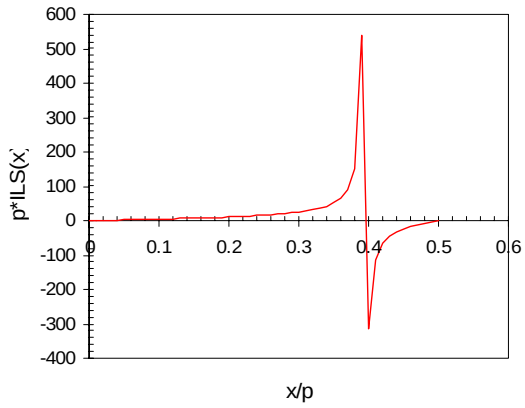
$$I(x) = a_0^2 + 4a_0a_1\cos(2\pi x/p) + 4a_1^2\cos^2(2\pi x/p)$$

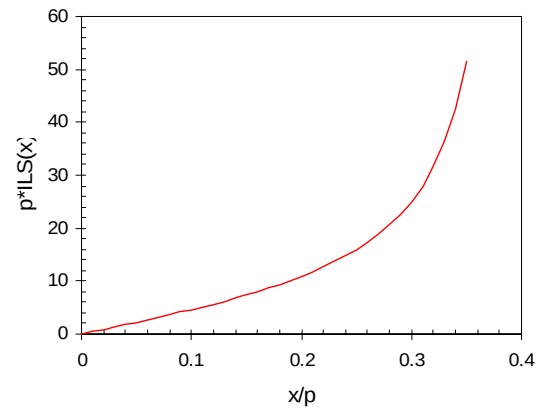
Assuming equal lines and spaces, plot pILS(x) versus x/p for $0 \le x \le p/2$.

For equal lines and spaces, $I(x) = \frac{1}{4} + \frac{2}{\pi}\cos(2\pi x/p) + \frac{4}{\pi^2}\cos^2(2\pi x/p)$ or

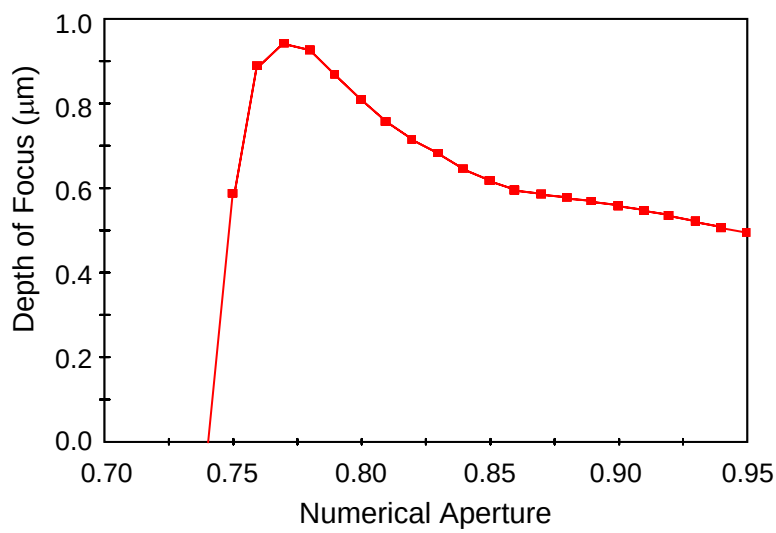
$I(x) = \frac{1}{4} + \frac{2}{\pi^2} + \frac{2}{\pi}\cos(2\pi x/p) + \frac{2}{\pi^2}\cos(4\pi x/p)$.

$-\frac{dI}{dx} = \frac{4}{p}\sin(2\pi x/p) + \frac{8}{\pi p}\sin(4\pi x/p)$, $ILS(x) = -\frac{1}{I}\frac{dI}{dx}$ (the minus sign is chosen to make the ILS positive for $0 \le x \le p/2$).





9.3. A plot of DOF versus NA (for a given feature) looks something like



Explain qualitatively why there is an optimum NA that maximizes the depth of focus.

For low NA, insufficient number of diffractive orders are making it through the lens and thus the image is not resolved. An unresolved image has zero depth of focus. Thus, at low NA (below the optimum NA), the image is resolution-limited and increasing the NA will increase DOF by allowing sufficient diffraction information to get a good enough quality image to be formed. For very high NA, the extra orders that enter the lens are at very high spatial frequencies and are thus very sensitive to

focus errors. At these NAs the image is focus-limited, meaning the DOF is limited by the sensitivity of the high spatial frequency components of the image. Between the resolution-limited regime and the focus-limited regime there is an optimum NA.

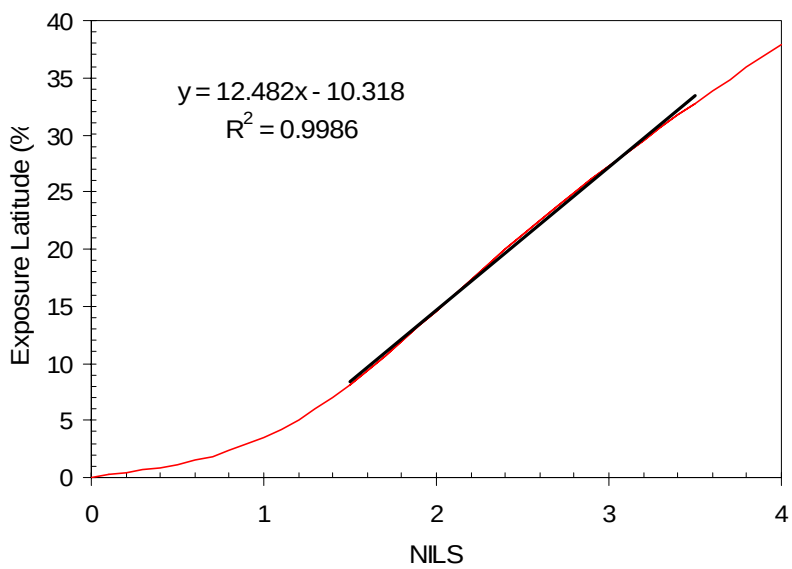
9.4. Using the following approximation for exposure latitude,

$$EL = 20\frac{\partial \ln E}{\partial \ln CD}$$

graphically compare equations (9.6) and (9.85) [let $\gamma = 10$ and $D_{eff}/CD = 2$] by plotting exposure latitude versus NILS. What are the best-fit values of α and β using NILS in the range from 1.5 to 3.5?

$$\%Exposure\ Latitude \approx \alpha(NILS - \beta) \quad \text{versus} \quad EL = \frac{20}{\frac{2}{NILS}\left(1+\frac{2}{\gamma\, NILS}\right)+\frac{2\gamma D_{eff}}{CD}e^{-\gamma\, NILS/4}} \quad \text{or}$$

$$EL = \frac{10NILS}{1+\frac{2}{\gamma\, NILS}+\gamma\, NILS\frac{D_{eff}}{CD}e^{-\gamma\, NILS/4}}$$



Thus, $\alpha = 12.5$, $\beta = 0.83$.

9.5. Using equations (9.20) and (9.21),

a) Derive an equation for $\left.\frac{\partial m^*/\partial x}{\partial m/\partial x}\right|_{x=p/4}$ for $N = 4$.

b) For this case, derive an expression for the approximate magnitude of the relative error in equation (9.22).

$$m(x)=\sum_{n=0}^{N}a_n\cos(2\pi nx/p)\quad\text{and}\quad m^*(x)=\sum_{n=0}^{N}a_n^*\cos(2\pi nx/p),\ \text{where}\ a_n^*=a_n e^{-2(\pi n\sigma_D/p)^2}$$

a) $$\left.\frac{\partial m^*/\partial x}{\partial m/\partial x}\right|_{x=p/4}=\left.\frac{\sum_{n=1}^{4}na_n^*\sin(2\pi nx/p)}{\sum_{n=1}^{4}na_n\sin(2\pi nx/p)}\right|_{x=p/4}=\frac{\sum_{n=1}^{4}na_n^*\sin(\pi n/2)}{\sum_{n=1}^{4}na_n\sin(\pi n/2)}=\frac{a_1^*-3a_3^*}{a_1-3a_3}$$

b) $$\frac{a_1^*-3a_3^*}{a_1-3a_3}=\left(\frac{a_1^*}{a_1}\right)\left(\frac{1-3\dfrac{a_3^*}{a_1^*}}{1-3\dfrac{a_3}{a_1}}\right).$$ For small a_3/a_1, $$\frac{a_1^*-3a_3^*}{a_1-3a_3}\approx\left(\frac{a_1^*}{a_1}\right)\left(1-3\frac{a_3^*}{a_1^*}\right)\left(1+3\frac{a_3}{a_1}\right)$$

$$\approx\left(\frac{a_1^*}{a_1}\right)\left(1+3\left[\frac{a_3}{a_1}-\frac{a_3^*}{a_1^*}\right]\right)=\left(\frac{a_1^*}{a_1}\right)\left(1+3\frac{a_3}{a_1}\left[1-\frac{e^{-18(\pi\sigma_D/p)^2}}{e^{-2(\pi\sigma_D/p)^2}}\right]\right)=\left(\frac{a_1^*}{a_1}\right)\left(1+3\frac{a_3}{a_1}\left[1-e^{-16(\pi\sigma_D/p)^2}\right]\right)$$

9.6. Derive equations (9.44) [hint: first assume m^* = constant, then assume h_{eff} is constant].

From $$\frac{\partial m^*}{\partial x}\approx\frac{1}{\eta}e^{-\alpha_f h_{eff}}\left(1-e^{-\eta\alpha_f}\right)\left(1-h_{eff}\right)\ln\left(1-h_{eff}\right)\frac{\partial\ln I}{\partial x},$$ derive

$$optimum\ \alpha_f=-\left[\frac{1+\ln\left(1-h_{eff}\right)}{\left(1-h_{eff}\right)\ln\left(1-h_{eff}\right)}\right]=\frac{1}{\eta}\ln\left(\frac{h_{eff}+\eta}{h_{eff}}\right)$$

First, assume $m^*=e^{-\alpha_f h_{eff}}$ = constant. Thus, we want to maximize $y=\left(1-e^{-\eta\alpha_f}\right)\left(1-h_{eff}\right)\ln\left(1-h_{eff}\right)$ where $1-h_{eff}=1+\dfrac{\ln(m^*)}{\alpha_f}$. So,

$$y=\left(1-e^{-\eta\alpha_f}\right)\left(1+\frac{\ln(m^*)}{\alpha_f}\right)\ln\left(1+\frac{\ln(m^*)}{\alpha_f}\right).\quad\frac{dy}{d\alpha_f}=0$$

$$0=\left(1-e^{-\eta\alpha_f}\right)\left(-\frac{\ln(m^*)}{\alpha_f^2}\right)\left(1+\ln\left(1+\frac{\ln(m^*)}{\alpha_f}\right)\right)+\left(1+\frac{\ln(m^*)}{\alpha_f}\right)\ln\left(1+\frac{\ln(m^*)}{\alpha_f}\right)\left(\eta e^{-\eta\alpha_f}\right)$$

$$-\frac{h_{eff}}{\alpha_f}\left(1-e^{-\eta\alpha_f}\right)\left(1+\ln\left(1-h_{eff}\right)\right)=\left(1-h_{eff}\right)\ln\left(1-h_{eff}\right)\left(\eta e^{-\eta\alpha_f}\right)$$

$$-\frac{1+\ln\left(1-h_{eff}\right)}{\left(1-h_{eff}\right)\ln\left(1-h_{eff}\right)}=\left(\frac{\eta\alpha_f}{h_{eff}}\right)\left(\frac{e^{-\eta\alpha_f}}{1-e^{-\eta\alpha_f}}\right)$$

Second, assume h_{eff} constant. Thus, we want to maximize $y=e^{-\alpha_f h_{eff}}\left(1-e^{-\eta\alpha_f}\right)$. $\dfrac{dy}{d\alpha_f}=0$,

$$0 = e^{-\alpha_f h_{eff}}\left(\eta e^{-\eta\alpha_f}\right) - \left(1 - e^{-\eta\alpha_f}\right)h_{eff}e^{-\alpha_f h_{eff}}, \quad \eta e^{-\eta\alpha_f} = \left(1 - e^{-\eta\alpha_f}\right)h_{eff}$$

$$optimum\ h_{eff} = \eta\left(\frac{e^{-\eta\alpha_f}}{1 - e^{-\eta\alpha_f}}\right).$$

Using this result above, we obtain $optimum\ \alpha_f = -\left[\frac{1 + \ln\left(1 - h_{eff}\right)}{\left(1 - h_{eff}\right)\ln\left(1 - h_{eff}\right)}\right]$.

Also, from $h_{eff} = \eta\left(\frac{e^{-\eta\alpha_f}}{1 - e^{-\eta\alpha_f}}\right)$ we can get $h_{eff}\left(1 - e^{-\eta\alpha_f}\right) = \eta e^{-\eta\alpha_f}$ and

$h_{eff} = \left(\eta + h_{eff}\right)e^{-\eta\alpha_f}$, $e^{\eta\alpha_f} = \left(\frac{\eta + h_{eff}}{h_{eff}}\right)$ and $optimum\ \alpha_f = \frac{1}{\eta}\ln\left(\frac{h_{eff} + \eta}{h_{eff}}\right)$.

9.7. Show that in the limit as $\eta \to 0$, equation (9.39) becomes equation (9.25).

$\frac{\partial m^*}{\partial x} \approx \frac{1}{\eta}m^*\left(1 - e^{-\eta\alpha_f}\right)\left(1 - h\right)\ln\left(1 - h\right)\frac{\partial \ln I}{\partial x}$. As $\eta \to 0$, $\lim_{\eta\to 0}\frac{1 - e^{-\eta\alpha_f}}{\eta} = \alpha_f$.
But, we also know that $\alpha_f = -\ln(m^*)/h$. Thus
$\frac{\partial m^*}{\partial x} \approx \alpha_f m^*\left(1 - h\right)\ln\left(1 - h\right)\frac{\partial \ln I}{\partial x} = -m^*\ln(m^*)\left(\frac{\left(1 - h\right)\ln\left(1 - h\right)}{h}\right)\frac{\partial \ln I}{\partial x}$. QED.

9.8. For the blocked polymer latent image when quencher is used, show that the maximum relative LIG becomes

$$\left.\frac{\partial m^*/\partial x}{ILS}\right|_{\max} \approx \frac{1}{\eta}(1 - q_0)\ln(1 - q_0)$$

whenever $\eta < \eta_c$.

Below the critical value of η, the optimum solution puts $h_{eff} = 0$ at the line edge. Thus
$\frac{\partial m^*}{\partial x} \approx \frac{1}{\eta}e^{-\alpha_f h_{eff}}\left(1 - e^{-\eta\alpha_f}\right)\left(1 - h_{eff} - q_0\right)\ln\left(1 - h_{eff} - q_0\right)\frac{\partial \ln I}{\partial x}$ becomes
$\left.\frac{\partial m^*/\partial x}{ILS}\right|_{\max} \approx \frac{1}{\eta}\left(1 - e^{-\eta\alpha_f}\right)(1 - q_0)\ln(1 - q_0)$. But, this solution also drives up the optimum amplification factor:
$optimum\ \alpha_f = \frac{1}{\eta}\ln\left(\frac{h_{eff} + \eta}{h_{eff}}\right)$, so at this optimum $1 - e^{-\eta\alpha_f} = 1 - \frac{h_{eff}}{h_{eff} + \eta} = \frac{\eta}{h_{eff} + \eta} \approx 1$ as h_{eff} goes to zero. Thus, $\left.\frac{\partial m^*/\partial x}{ILS}\right|_{\max} \approx \frac{1}{\eta}(1 - q_0)\ln(1 - q_0)$. QED.

9.9. Show that at the optimum conditions (i.e., when the blocked polymer LIG is at its maximum),

$$\frac{1-e^{-\alpha_f \eta}}{\eta} = \frac{1}{h_{eff}+\eta} \quad \text{and} \quad optimum\ h_{eff} = \eta\left(\frac{e^{-\alpha_f \eta}}{1-e^{-\alpha_f \eta}}\right)$$

We know that $optimum\ \alpha_f = \frac{1}{\eta}\ln\left(\frac{h_{eff}+\eta}{h_{eff}}\right)$, so $e^{-\eta\alpha_f} = \frac{h_{eff}}{h_{eff}+\eta}$. Thus,

$1-e^{-\alpha_f \eta} = 1-\frac{h_{eff}}{h_{eff}+\eta} = \frac{\eta}{h_{eff}+\eta}$ and $\frac{1-e^{-\alpha_f \eta}}{\eta} = \frac{1}{h_{eff}+\eta}$. Solving this equation for h_{eff} will give the optimum value of the effective acid concentration:

$$h_{eff}+\eta = \frac{\eta}{1-e^{-\alpha_f \eta}}, \quad h_{eff} = \frac{\eta}{1-e^{-\alpha_f \eta}} - \eta = \eta\left(\frac{1}{1-e^{-\alpha_f \eta}} - 1\right) = \eta\left(\frac{e^{-\alpha_f \eta}}{1-e^{-\alpha_f \eta}}\right).$$

9.10. Consider a resist obeying the original kinetic development model so that n = 12, m_{th} = 0.65, and r_{max}/r_{min} = 2,000. If one desired to reformulate the resist to increase the maximum log-development rate versus m^* by 20%, how much would each one of these variables have to be individually changed?

For large n,

$$-\left.\frac{\partial \ln r}{\partial m^*}\right|_{\max} \approx \frac{n}{(1-m_{th})} + \frac{1}{2(1-m_{th})}\ln\left(\frac{r_{\max}}{r_{\min}}\right) = \frac{12}{0.35} + \frac{1}{2(0.35)}\ln(2{,}000) = 34.3 + 10.9 = 45.1$$

To increase this value to 54.2:

$$54.2 = \frac{n}{0.35} + 10.9, \quad n = 15.1$$

$$54.2 = \frac{12}{(1-m_{th})} + \frac{7.6}{2(1-m_{th})}, \quad m_{th} = 0.71$$

$$54.2 = \frac{12}{0.35} + \frac{1}{2(0.35)}\ln\left(\frac{r_{\max}}{r_{\min}}\right), \quad \frac{r_{\max}}{r_{\min}} = 1{,}120{,}000$$

9.11. In the limit of $\sigma_D \gg a$, derive an approximate proportionality for LER. In this regime, how does increasing diffusion affect LER?

$$LER \propto \frac{\sigma_D^{\ 2}}{1-e^{-\pi^2\sigma_D^{\ 2}/2L^2}}\sqrt{\frac{\langle m^*\rangle}{\langle n_{0-blocked}\rangle} + \left(\langle m^*\rangle \ln\langle m^*\rangle\right)^2\left(\frac{a}{\sigma_D}\right)^p\left(\frac{1}{\langle n_{0-PAG}\rangle\langle h\rangle}\right)}$$

For $\sigma_D \gg a$,

$$LER \propto \frac{\sigma_D^{\ 2}}{1-e^{-\pi^2\sigma_D^{\ 2}/2L^2}}\sqrt{\frac{\langle m^*\rangle}{\langle n_{0-blocked}\rangle}} \quad \text{if} \quad -(K_{amp}t_{PEB})\langle m^*\rangle\ln\langle m^*\rangle\left(\frac{a}{\sigma_D}\right)^p\frac{\langle n_{0-block}\rangle}{\langle n_{0-PAG}\rangle} << 1.$$

Using typical numbers, $K_{amp} t_{PEB} = 2$, $-\langle m^* \rangle \ln \langle m^* \rangle = 0.3$, $\frac{\langle n_{0-block} \rangle}{\langle n_{0-PAG} \rangle} = 29$, $p = 3$, the requirement is that $\frac{\sigma_D}{a} >> 2.6$.

In this regime, diffusion will always make LER worse.

Chapter 10

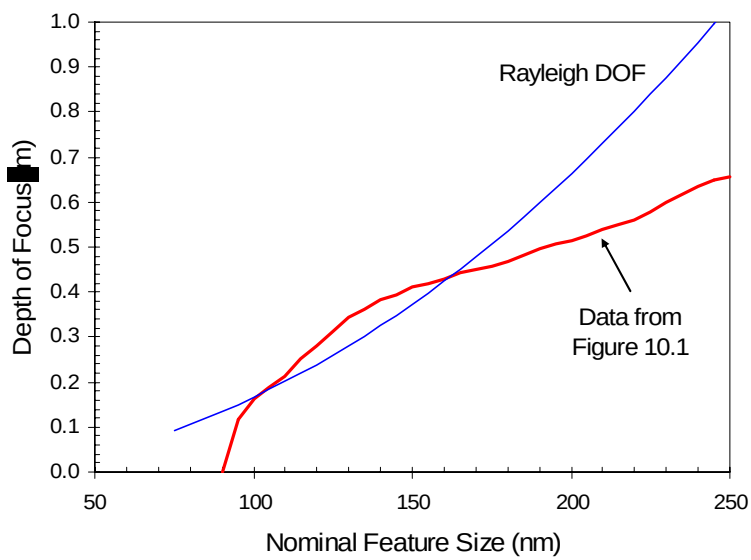
10.1. Figure 10.1 shows an example of how the depth of focus varies with feature size. Generate an equivalent plot using the Rayleigh DOF criterion. Assume equal lines and spaces, coherent three-beam imaging , and $k_2 = 0.8$.

The Rayleigh DOF criterion applies the features at the resolution limit. For coherent three-beam imaging, this occurs at a feature size of $w = k_1\lambda/NA$ where $k_1 = 0.5$. Thus, the Rayleigh DOF criterion can be written as

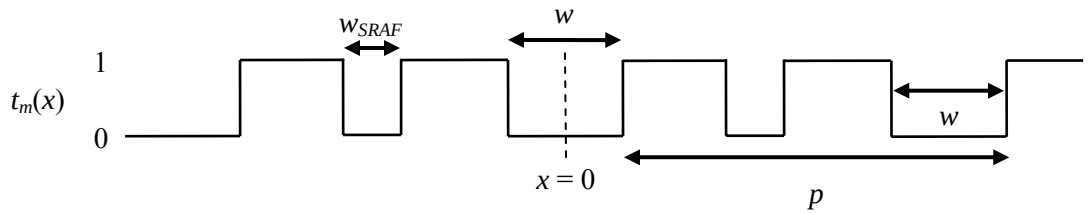
$$DOF = k_2 \frac{\lambda}{NA^2} = \frac{k_2}{k_1}\frac{w}{NA} = \frac{k_2}{k_1^2}\frac{w^2}{\lambda}$$

Thus, we have two options: Assume NA is constant (so that the wavelength is varied in order to keep the feature size at the resolution limit), or constant wavelength so that NA is varied. The second option is the more reasonable, since most lithography tools have variable NA but a fixed wavelength. To generate the plot, we must assume a wavelength – let's pick 193 nm. Thus,

$$DOF = \frac{k_2}{k_1^2}\frac{w^2}{\lambda} = \frac{w^2}{60.3nm}$$



10.2. Derive the diffraction pattern of a semi-dense line (width w, pitch p) with single SRAFs (width w_{SRAF}) placed in the center of the spaces:



The diffraction pattern for a repeating pattern of spaces of width w and pitch p, centered at $x = 0$:

$$T_m(f_x) = \frac{1}{p}\sum_{j=-\infty}^{\infty} \frac{\sin(\pi f_x w)}{\pi f_x}\delta\left(f_x - \frac{j}{p}\right)$$

The diffraction pattern of a second repeating pattern of spaces of width w_{SRAF} and pitch p, shifted by $p/2$:

$$T_m(f_x) = -\frac{1}{p}\sum_{j=-\infty}^{\infty} \frac{\sin(\pi f_x w_{SRAF})}{\pi f_x}\delta\left(f_x - \frac{j}{p}\right)e^{-i\pi f_x p}$$

Using superposition to add these two arrays together:

$$T_m(f_x) = \frac{1}{p}\sum_{j=-\infty}^{\infty} \frac{1}{\pi f_x}\delta\left(f_x - \frac{j}{p}\right)\left(\sin(\pi f_x w) + \sin(\pi f_x w_{SRAF})e^{-i\pi f_x p}\right)$$

Since the pattern is lines rather than spaces, $T_{line}(f_x) = \delta(f_x) - T_{space}(f_x)$ giving the final result:

$$T_m(f_x) = \delta(f_x) - \frac{1}{p}\sum_{j=-\infty}^{\infty} \frac{1}{\pi f_x}\delta\left(f_x - \frac{j}{p}\right)\left(\sin(\pi f_x w) + \sin(\pi f_x w_{SRAF})e^{-i\pi f_x p}\right)$$

10.3. Considering the FEM plots of Figure 10.14 to be typical of dense and isolated lines, how does the iso-dense bias vary with focus?

For the dense features (figure 10.14a) , the isofocal bias is about zero. Thus, the dense CD is about independent of focus. The isolated CD, however, decreases as it goes out of focus (Figure 10.14b). At best focus and the nominal dose (which looks to be about 11 mJ/cm^2 from the figures), the iso-dense bias is about zero. Out of focus, the iso-dense bias goes negative, reaching -130 nm at the extremes of focus.

10.4. What is the smallest value of k_{pitch} possible under the following scenarios:
a) COG mask and conventional illumination as σ goes to zero (i.e., coherent illumination)
b) Alt-PSM mask and conventional illumination as σ goes to zero (i.e., coherent illumination)
c) COG mask and dipole illumination with very small poles
d) COG mask and quadrupole illumination with very small poles

a) This is coherent three-beam imaging, so the smallest pitch would put the first diffracted order at the edge of the pupil. Thus, the minimum $k_{pitch} = 1$.
b) For alt-PSM (pure two-beam imaging), the minimum pitch is cut in half compared to coherent three-beam imaging and the minimum $k_{pitch} = 0.5$.
c) This scenario also results in pure two-beam imaging, so the minimum $k_{pitch} = 0.5$.
d) Quadrupole illumination gives two-beam imaging, but the diffraction orders are shifted up and down in the pupil so that the maximum distance between diffraction orders is not the diameter of the pupil, but the diameter divided by √2. Thus, the minimum k_{pitch} is increased by this amount compared to dipole illumination and the minimum $k_{pitch} = 0.707$.

10.5. Explain the main advantage of dipole illumination compared to quadrupole illumination. Explain the main disadvantage of dipole illumination compared to quadrupole illumination.

Both dipole and quadrupole illumination give two-beam imaging, but for quadrupole illumination the diffraction orders are shifted up and down in the pupil so that the maximum distance between diffraction orders is not the diameter of the pupil, but the diameter divided by √2. Thus, the minimum k_{pitch} is increased by this amount compared to dipole illumination:

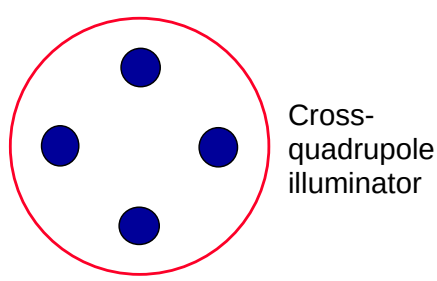
For dipole illumination: minimum $k_{pitch} = 0.5$.
For quadrupole illumination: minimum $k_{pitch} = 0.707$.

Thus, the main advantage of dipole illumination is improved pitch resolution. This improved resolution only occurs for one orientation of lines and spaces, however (either vertical patterns or horizontal patterns, but not both). Quadrupole, on the other hand, can image vertical and horizontal features equally well. Thus, the main disadvantage of dipole illumination is that it restricts the design to use only one orientation of critical features.

10.6. For an optimized dipole illuminator, a two-beam aerial image results:

$$I(x) = a_0^2 + a_1^2 + 2a_0a_1\cos(2\pi x/p)$$

Since the dipole can only print small-pitch patterns oriented in one direction, one possible solution is to add a second dipole rotated by 90° to obtain an illuminator called a *cross-quadrupole*:



When printing at the pitch for which the dipole and quadrupole are optimized, derive the resulting in-focus aerial image (assume TE illumination) for a cross-quadrupole.

Assuming vertically oriented lines and spaces, the right and left poles will produce the standard two beam image. The top and bottom poles, on the other hand, will produce a one-beam image (constant

intensity pattern. Thus, when all poles are included, the resulting image will be the average of the one-beam and two-beam images:

$$I_{1-beam} = a_0^2$$
$$I_{2-beam} = a_0^2 + a_1^2 + 2a_0 a_1 \cos(2\pi x/p)$$
$$I_{cross-quad} = 0.5\left(I_{1-beam} + I_{2-beam}\right) = a_0^2 + 0.5a_1^2 + a_0 a_1 \cos(2\pi x/p)$$

10.7. The coherent two-beam aerial image for an alternating PSM mask is given in equation (10.9). From this, derive the image for a chromeless repeating line/space pattern by letting the width of the chrome line in the alternating PSM mask go to zero. Qualitatively, what is the effect of reducing the chrome width?

Equation (10.9) shows $I(x) = 2a_1{}^2\left(1 + \cos(2\pi x/p)\right)$, $a_1 = \dfrac{2\sin\left(\pi w/2p\right)}{\pi}$ where w is the width of the space. Letting the width of the line go to zero, the space width becomes equal to p, so that $a_1 = 2/\pi$. The resulting aerial image is $I(x) = \dfrac{8}{\pi^2}\left(1 + \cos(2\pi x/p)\right)$. Thus, the only impact of reducing the chrome width (for pitches small enough to generate two-beam images) is a scaling of the image intensity – a reduction of the dose needed to image the pattern, but no change in the quality of the pattern.

10.8. Derive equation (10.10), the diffraction pattern for an isolated phase-shifted line.

Setting up the diffraction pattern as the Fourier transform of the mask transmittance,

$$T_m(f_x) = \int_{-\infty}^{-w/2} e^{-i2\pi f_x x} dx - \int_{w/2}^{\infty} e^{-i2\pi f_x x} dx$$

To make things easier, shift each integral's x-axis by $w/2$ or $-w/2$:

$$T_m(f_x) = e^{-i\pi f_x w} \int_{-\infty}^{0} e^{-i2\pi f_x x} dx - e^{+i\pi f_x w} \int_{0}^{\infty} e^{-i2\pi f_x x} dx$$

Now, each integral is just the diffraction pattern of an isolated edge. Using the table of Fourier transforms from Chapter 2,

$$T_m(f_x) = e^{-i\pi f_x w}\left[\frac{1}{2}\delta\left(f_x\right) + \frac{i}{2\pi f_x}\right] - e^{+i\pi f_x w}\left[\frac{1}{2}\delta\left(f_x\right) - \frac{i}{2\pi f_x}\right]$$

$$T_m(f_x) = i\frac{\cos\left(\pi f_x w\right)}{\pi f_x} - i\sin\left(\pi f_x w\right)\delta\left(f_x\right)$$

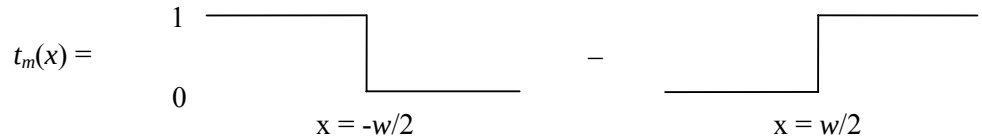
However, evaluating the sine at $f_x = 0$, the only place where the delta function is non-zero, gives

$$T_m(f_x) = i\frac{\cos\left(\pi f_x w\right)}{\pi f_x}$$

10.9. Derive equation (10.11), the coherent aerial image for an isolated phase-shifted line.

A simple approach is to consider the mask pattern as the superposition of two isolated binary edges:

Or



The electric field image of an isolated edge is (see homework problem 2.21):

$$E_{edge}(x) = \frac{1}{2} + \frac{1}{\pi} Si\left(\frac{2\pi NAx}{\lambda}\right)$$

The total electric field, using superposition, is then

$$E(x) = E_{edge}(-x - w/2) - E_{edge}(x - w/2) = -\frac{1}{\pi}\left[\mathrm{Si}\left(\frac{2\pi NA}{\lambda}(x + w/2)\right) + \mathrm{Si}\left(\frac{2\pi NA}{\lambda}(x - w/2)\right)\right].$$

(Remember that the Sine Integral, Si, is an odd function.)

Thus, $I(x) = \frac{1}{\pi^2}\left[\mathrm{Si}\left(\frac{2\pi NA}{\lambda}(x + w/2)\right) + \mathrm{Si}\left(\frac{2\pi NA}{\lambda}(x - w/2)\right)\right]^2$

10.10. Derive equation (10.15) from equation (10.14).

Equation (10.14): $a_j = t_1 e^{i\phi_1}\left(\frac{\sin(j\pi w_1/p)}{j\pi}\right) + t_2 e^{i\phi_2}(-1)^j\left(\frac{\sin(j\pi w_2/p)}{j\pi}\right)$

Consider $j = 0$. Since $\sin(x)/x$ goes to 1 as x goes to 0,

$a_0 = t_1 e^{i\phi_1}\left(\frac{w_1}{p}\right) + t_2 e^{i\phi_2}\left(\frac{w_2}{p}\right)$. But, since $w_1 + w_2 = p$,

$$a_0 = t_1 e^{i\phi_1}\left(\frac{w_1}{p}\right) + t_2 e^{i\phi_2}\left(\frac{p - w_1}{p}\right) = \left(t_1 e^{i\phi_1} - t_2 e^{i\phi_2}\right)\left(\frac{w_1}{p}\right) + t_2 e^{i\phi_2}$$

For j not equal to zero:

$\sin(j\pi w_2/p) = \sin(j\pi[p - w_1]/p) = \sin(j\pi - j\pi w_1/p) = -(-1)^j \sin(j\pi w_1/p)$. Thus,

$$a_j = \left(t_1 e^{i\phi_1} - t_2 e^{i\phi_2}\right)\left(\frac{\sin(j\pi w_1/p)}{j\pi}\right)$$

10.11. Equations (10.16) and (10.17) give the coherent (TE) image for an attenuated PSM pattern of lines and spaces. What happens to this image as t_2 goes to 1? (Such a mask is called a chromeless PSM.)

Equation (10.16): $a_0 = (1+t_2)\left(\frac{w_1}{p}\right) - t_2$, $a_j = a_{-j} = (1+t_2)\left(\frac{\sin(j\pi w_1/p)}{j\pi}\right)$.

Equation (10.17): $I(x) = a_0^2 + 2a_1^2 + 4a_0a_1\cos(2\pi x/p) + 2a_1^2\cos(4\pi x/p)$.

Letting t_2 go to 1, $a_0 = 2\left(\frac{w_1}{p}\right) - 1$, $a_1 = \frac{2\sin(\pi w_1/p)}{\pi}$. For the case of equal lines and spaces, w_1/p = 0.5, so that a_0 = 0. Thus, for this case, the image will become a two-beam image rather than a three-beam image.

10.12. Derive an equation for the coherent LSF when using an attenuated PSM. What is the maximum sidelobe intensity for the LSF for a 6% attenuating PSM?

For a binary mask, $LSF(x) = \left[\frac{\sin(2\pi NAx/\lambda)}{2\pi NAx/\lambda}\right]^2$. By analogy with the PSM for an attenuated mask, the resulting attenuated PSM LSF will be

$$LSF(x) = \left[\frac{\frac{\sin(2\pi NAx/\lambda)}{2\pi NAx/\lambda} - t}{1-t}\right]^2.$$

Numerically evaluating this expression for t = sqrt(0.06), the peak sidelobe intensity will be 0.3747.

10.13. Derive an expression for the normalized image log-slope of the isolated phase-edge, assuming the nominal feature size is at an intensity threshold of 0.25.

For coherent TE illumination, the isolated phase edge image will be $I(x) = \frac{4}{\pi^2}\mathrm{Si}^2\left(\frac{2\pi NAx}{\lambda}\right)$. Taking the derivative, $\frac{dI}{dx} = \frac{8}{\pi^2}\mathrm{Si}\left(\frac{2\pi NAx}{\lambda}\right)\left(\frac{\sin(2\pi NAx/\lambda)}{x}\right)$. Evaluating the image at the nominal feature edge, $I(x_{edge}) = \frac{4}{\pi^2}\mathrm{Si}^2\left(\frac{2\pi NAx_{edge}}{\lambda}\right) = 0.25$ gives $\mathrm{Si}\left(\frac{2\pi NAx_{edge}}{\lambda}\right) = \frac{\pi}{4}$ and $x_{edge} = 0.13\lambda/NA$.

Thus, $NILS = \frac{16}{\pi}\sin(0.26\pi) = 3.71$.